It Was Already Dobbs:
Abortion Bans and Maternal Health

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Clinicians warned that post-*Dobbs* bans would harm maternal health, but ban states had long restricted abortion, possibly limiting a ban's effect. Using claims on nearly 1 million insured pregnancies and stacked difference-in-differences, I find that the bans changed maternal health and the timing and intensity of care little, while suggestively raising fertility and the live-birth share. Adverse pregnancy outcomes rose 2.8 percent, consistent with more deliveries, but intervals rule out increases above 4.9 percent, and 5.6 percent for severe maternal morbidity. Restoring a federal abortion right alone may not meaningfully improve insured women's maternal health.

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Between September 2021 and August 2023, fifteen U.S. states began enforcing bans on abortion at six weeks of gestation or earlier (Dench, Myers and Pineda-Torres, 2025, Appendix A). The Supreme Court enabled these laws with its June 2022 decision in *Dobbs v. Jackson Women's Health Organization*, which overturned the constitutional right to abortion recognized in *Roe v. Wade*. The decision was the largest change to abortion law in half a century, what Dench, Pineda-Torres and Myers (2024) call "the most profound transformation of the landscape of abortion access in 50 years." As the bans took effect, clinicians, medical societies, and journalists warned that the harm would reach beyond women seeking an elective abortion for reasons unrelated to their health (American College of Obstetricians and Gynecologists, 2022; Harris, 2022; Simmons-Duffin, 2022).

Three features of abortion care and of the bans drove this concern. First, the medications and procedures used to end a pregnancy are also the standard treatment for miscarriage, ectopic pregnancy, and several severe complications of pregnancy, so their provision could be mistaken for an abortion. Second, the line between an "elective" and a "medically necessary" abortion is often unclear. For instance, a pregnancy with a severe fetal anomaly may pose no imminent risk to the woman when the anomaly is diagnosed but can become riskier as it continues (Nambiar et al., 2025). Third, although the bans include exceptions for the woman's life and sometimes her health (Felix, Sobel and Salganicoff, 2024), they say little about how and when a physician should act in these ambiguous cases, and bans threaten physicians who violate them with severe penalties (Felix, Sobel and Salganicoff, 2025). Clinician case reports and investigative reporting have since documented women with serious pregnancy complications sent home from emergency departments until their condition deteriorated enough to meet an exception (Advancing New Standards in Reproductive Health, 2024; Seitz, 2024). The prevailing expectation was therefore that the bans would mark a sharp break in the care pregnant women received and in their health.

However, the abortion bans enabled by *Dobbs* followed a decade of tightening restrictions on abortion in the same states that would ultimately go on to ban the procedure. By the start of 2022, many states that would later ban abortion already imposed waiting periods and regulations targeting abortion providers that closed clinics and lengthened travel distances (Guttmacher Institute, 2022; Lindo et al., 2020). In 2021 alone, states enacted 108 new restrictions, more than in any earlier year since *Roe* (Nash, 2021). Further, many forms of insurance had stopped paying for most abortions long before any formal ban. The Hyde Amendment has barred federal Medicaid funds from paying for abortion in most circumstances since 1977 (Salganicoff et al., 2024), and many of these states also limited coverage in private plans and in plans sold on the Affordable Care Act Marketplace (Guttmacher Institute, 2021*a*). Even before *Dobbs*, women in states whose Medicaid

programs did not pay for abortion paid out of pocket for three in four of the abortions they obtained (Jerman, Jones and Onda, 2016), and among residents of states hostile to abortion rights, the share who traveled out of state for an abortion rose from 9 percent in 2011 to 15 percent in 2020 (Maddow-Zimet and Kost, 2022). In these states, access to abortion likely already resembled a post-*Dobbs* environment well before *Dobbs*.

How the bans changed the maternal health landscape is therefore an empirical question of great interest and significance. This paper provides the first comprehensive evidence on this question by estimating the effect of abortion bans on pregnancy endings, maternal health, and the timing and intensity of maternal health care. From 350 million insurance claim lines, I construct pregnancy episodes that follow each woman from a year before conception through the postpartum period and record the diagnoses, procedures, and providers for services billed to her insurance. My analysis sample follows the nearly 1 million pregnancies of women insured from 2020 through 2024. Using a stacked difference-in-differences design, I compare pregnancies in ban states with pregnancies in states that continuously protected abortion, before and after each state imposed its ban. I study three groups of outcomes. First, outcomes that describe the ending of a pregnancy (the shares ending in a live birth, a miscarriage, or an abortion billed to insurance, and the share with no observed ending) show whether the bans changed how pregnancies ended. Second, maternal health outcomes (a composite of adverse pregnancy outcomes, hypertensive disorders of pregnancy, preterm birth, and severe maternal morbidity) test the warnings of harm to women's health directly. Third, timing and intensity of care outcomes (the days to the first visit, the length of the delivery stay, the length of pregnancy, and the number of visits) describe the care women received. Finally, to understand why these outcomes moved as they did, I examine whether the bans changed fertility and the pre-pregnancy health of the women who became pregnant.

I find that *Dobbs*-enabled abortion bans led to minimal change in maternal health and in the timing and intensity of care for pregnant women, while modestly raising fertility and the share of pregnancies ending in a live birth. In my stacked design, I first find that the live-birth share rose a suggestive 2.7 percent relative to protected states. This shift helps explain the one maternal health outcome that rises. A composite of common adverse pregnancy outcomes, including infection, preterm birth, and gestational diabetes, is 2.8 percent higher ($p = 0.01$), with an interval that rules out increases larger than 4.9 percent. Because these outcomes are far more common in deliveries than in losses, the added live births alone account for most of this rise, and among live births the composite changes by -1.2 percent, ruling out increases larger than 0.6 percent. Changes in the most common complications are also small, as hypertensive disorders change by 1.0 percent and

preterm birth among live births by −0.9 percent, with intervals that rule out increases larger than 3.4 and 2.6 percent. Changes in severe maternal morbidity, a rare but serious outcome at the center of the post-*Dobbs* warnings, are also small, at −2.9 percent, ruling out increases larger than 5.6 percent. Lastly, the timing of care changed little, as the days to the first visit change by 1.4 percent, with an interval that rules out delays longer than 3.8 percent. However, these estimates average over all insured pregnancies, most of which are routine pregnancies carried to delivery. Among losses, the small group the warnings most concerned, severe maternal morbidity changes by 11.3 percent, but its interval reaches 33.4 percent, so I cannot rule out harm concentrated among the pregnancies whose complications call for urgent care or a termination.

These data also document a pre-ban landscape that places these estimates in context, and two facts are worth noting. First, abortion was already largely outside the insurance system in these states. In 2020, before any formal ban, insurance-billed abortion ended about 1 percent of ban-state pregnancies with an observed ending, against 14 percent of protected-state pregnancies. Second, maternal health in ban states was already worse, as pregnant women in ban states were more often diagnosed with a hypertensive disorder or an adverse pregnancy outcome such as infection or gestational diabetes, and even their live births more often ended preterm. These gaps persist after adjusting for women's demographic and health characteristics. To my knowledge, these facts are among the first to document this landscape at this level of detail, for every insured pregnancy rather than only for births. Claims make this possible, because they follow a pregnancy from start to ending, while birth certificates and abortion surveillance records each see only some endings.

Turning to the fertility channel, I find suggestive evidence that fertility rose. The birth rate among enrolled women is 3.7 percent higher relative to protected states ($p = 0.02$), with positive estimates in every subgroup, including 5.5 percent among women with a high-risk condition, although the subgroup intervals overlap. The pregnancy rate changes little, so the added births appear to come from more pregnancies reaching a live birth rather than from more conceptions, in line with the rise in the live-birth share. Despite this rise in births, the women who became pregnant after the bans were not measurably less healthy, as the share entering pregnancy with any condition in the obstetric comorbidity index changed by 0.4 percent, ruling out increases larger than 2.6 percent.

I contribute to two active streams of literature. The first studies how abortion restrictions shape maternal health. Studies of the legalization of abortion around the time of *Roe* find that it lowered maternal mortality, by 30 to 50 percent among nonwhite women (Farin, Hoehn-Velasco and Pesko, 2024), and studies of later restrictions find that they worsened maternal health. Regulations targeting abortion providers raised hypertensive disorders of pregnancy (Gardner, 2024). In Texas

after its 2021 ban, sepsis rose among women hospitalized with a second-trimester loss (Presser et al., 2025) and critical illness among high-risk hospitalized pregnancies (Chen et al., 2026), in before-and-after comparisons within the state, and blood transfusions for pregnancy loss rose after its 2022 total ban (Nagle et al., 2025). Women denied a wanted abortion also report worse physical health and more financial distress in the years that follow (Ralph et al., 2019; Miller, Wherry and Foster, 2023). In Mexico, by contrast, harsher penalties in states where abortion was already illegal had small or null effects on women's health (Clarke and Mühlrad, 2021). Early evaluations of *Dobbs* find that pregnancy-associated mortality may have increased, although the estimates have wide intervals (Bell et al., 2026; Abraha et al., 2026), while others find that the gap between ban states and other states changed little once pandemic-era deaths are accounted for (Stevenson and Root, 2024; Nuss et al., 2025). Infant deaths rose, including deaths from congenital anomalies (Gemmill et al., 2024, 2025), and most among Black infants in Texas (Caraher, 2024). Closest to my design, a difference-in-differences study of commercial claims in 14 trigger-ban states and 17 others finds that the bans shifted miscarriage care toward expectant management, by 2.8 percentage points (Rodriguez, Fuerst and Schrote, 2026). I extend this literature beyond the management of one type of loss to a broad set of health outcomes and pregnancy endings, measured in detailed claims that describe care from conception through the postpartum period. To my knowledge, this is the first comprehensive evidence on the effect of the post-*Dobbs* bans on these outcomes in a large sample of insured pregnancies. However, in contrast to some of this work, I find that the bans enabled by *Dobbs* changed maternal health little, a difference that my broader sample, which averages over all insured pregnancies rather than high-risk hospitalizations, may explain.

The second literature measures the fertility response to abortion restrictions. It generally finds that restrictions raise births modestly (Fischer, Royer and White, 2018; Venator and Fletcher, 2021; Bell, Stuart and Gemmill, 2023; Caraher, 2024), and that the bans after *Dobbs* raised them by about 2 percent (Dench, Pineda-Torres and Myers, 2024; Bell et al., 2025). Restrictions short of a ban had already lowered abortion rates, most of all those aimed at the women seeking care rather than at providers (Caraher, 2026). Unlike the birth certificates these studies use, my sample excludes uninsured women and women whose coverage changed between 2020 and 2024. In this sample, I find modest and suggestive evidence of a rise in births that partly continues earlier trends. Because claims also record pregnancies that do not end in a birth, I can measure the pregnancy rate, which birth certificates cannot, and separate pregnancies carried to term from added conceptions. Birth certificates record only a few conditions before conception, and only for pregnancies that end in a birth, while claims record every diagnosis a woman receives in the year before she conceives.

I can therefore test whether the bans changed who became pregnant, as Romania's 1966 abortion ban did (Pop-Eleches, 2006).

I confront several threats to the validity of my design. First, women enter and leave my data for many reasons, some of which may correlate with treatment. Therefore, I restrict my analysis sample to women continuously enrolled from 2020 through 2024, to keep a stable denominator and to follow women before and after pregnancy (Section II.D). However, this requirement may select women better able to circumvent a ban, for example by traveling, and such selection would lead the estimates to understate rather than overstate the effect of the bans. To check this restriction, in Supplemental Appendix A4 I use a weaker enrollment requirement and find similar health estimates, although the shift in endings there runs toward miscarriages rather than live births. I further show that the pregnancies in my sample broadly resemble all U.S. births demographically (Supplemental Appendix Table A7). Second, protected states expanded access to abortion over the same years that ban states continued to restrict it, so their paths before and after the bans could differ from those of ban states. To address this concern, I reweight protected states to match the pre-ban paths of ban states with synthetic difference-in-differences (Supplemental Appendix A8), and in Supplemental Appendix A9 I compare women near and far from protected-state care within ban states, a comparison that leaves out protected states entirely. Both designs give maternal health estimates close to or below the main ones, with wider intervals that include zero. Lastly, the bans may change who becomes pregnant. Because claims record each woman's health before conception, I measure this directly and find little evidence that changing composition masks harm (Section VI.B).

Taken together, my results place the *Dobbs* bans in the environment of restrictions that preceded them. Maternal health in ban states was already worse than in protected states before the bans and changed little after them, while fertility and the share of pregnancies ending in a live birth rose modestly. These patterns are consistent with the earlier restrictions having already done much of what a ban could do to maternal health. For instance, Gardner (2024) finds that targeted regulations of abortion providers, which most later ban states had adopted, raised hypertensive disorders of pregnancy by 11 to 16 percent, while the 95 percent confidence interval for the effect of the bans on the same outcome runs from -1.5 to 3.4 percent. Difference-in-differences studies also find that the same regulations had already lowered the supply of obstetrician-gynecologists by about 5 to 7 percent (Chatterji, Ho and Qi, 2026; Vandenbroeck and Markowski, 2025). My design cannot fully separate this explanation from another, that insured women in ban states got around the bans by traveling or by telehealth, but under either explanation a ban added little harm to maternal health beyond what these women already faced. However, my estimates bound

how common severe harm concentrated among the few percent of pregnancies with the hardest-to-treat complications could have been, not whether it occurred. Finally, these findings suggest that returning the federal right to abortion alone may not meaningfully change the maternal health and care landscape for insured women.

The paper proceeds as follows. Section I describes the history of restrictions before and after *Dobbs* and how abortion restrictions could affect maternal health. Section II describes the data, samples, and outcomes, Section III the pre-ban landscape, and Section IV the empirical strategy. Section V presents the effects on pregnancy endings, maternal health, and the timing and intensity of care, and Section VI the fertility channel. Section VII concludes.

I. Background and Expected Effects

A. Abortion Policy Before and After Dobbs

The history of abortion restrictions in states that enacted post-*Dobbs* abortion bans began long before *Dobbs*, and the restrictions grew tighter in the years leading up to the decision (Nash, 2021). By the start of 2022, every one of the fifteen states that would later ban abortion already imposed a gestational limit on abortion and made women wait 18 to 72 hours between counseling and the procedure (Guttmacher Institute, 2022; *Bristol Regional Women's Center, P.C. v. Slatery*, 2021).[1] Eleven states enforced targeted regulations of abortion providers, rules that go beyond what patient safety requires, often holding clinics to the standards of ambulatory surgical centers (Guttmacher Institute, 2021*b*). Studies of these rules find that provider regulations closed clinics and lengthened the distances women traveled to reach a provider (Grossman et al., 2017; Lindo et al., 2020), and that a waiting period pushed abortions later in pregnancy (Lindo and Pineda-Torres, 2021). Some rules already carried criminal penalties for physicians, as in Texas, where an abortion without the required sonogram had been a crime since 2011 (Texas State Law Library, 2026).

Insurance coverage was minimal in states that ultimately banned abortion as well. Beyond the federal Hyde Amendment's limits on Medicaid, none of the fifteen states with post-*Dobbs* bans used their own funds to pay for most medically necessary abortions for women on Medicaid (Guttmacher Institute, 2022), and fourteen of the fifteen also restricted abortion coverage in plans sold on the Affordable Care Act Marketplace. Six restricted it in every private plan written in the state (Guttmacher Institute, 2021*a*). As a result, insurance pays for only a minority of abortions in the United States (Roberts et al., 2014). Coverage also matters for access, because in 2020 a patient

[1] Guttmacher's January 2022 table lists Tennessee's 48-hour waiting period as enjoined, but the Sixth Circuit allowed it to take effect in April 2021 and upheld it in August 2021 (*Bristol Regional Women's Center, P.C. v. Slatery*, 2021).

paying out of pocket faced a median charge of $560 for a medication abortion (Upadhyay et al., 2022). Previous work has found that state restrictions on Medicaid funding reduced abortions, with the response concentrated among low-income women (Levine, Trainor and Zimmerman, 1996), and a review of this literature concludes that about one in four women who would have had a Medicaid-funded abortion give birth when that funding is unavailable (Henshaw et al., 2009).

In addition, several of these states had already passed near-total or early bans, only to see them enjoined under *Roe*, so physicians potentially already practiced under laws that could change from year to year, and potentially shifted their practice patterns to adapt to strict environments, whether or not the laws were enforced. For example, Georgia's six-week ban passed in 2019 but took effect only after *Dobbs* (*SisterSong v. Governor of Georgia*, 2022), and legal challenges kept similar bans in Arkansas, Kentucky, and Ohio from taking effect (Nash et al., 2019; DeMillo, 2021). Even restrictions that courts struck down could leave lasting effects. In Texas, a 2013 law requiring abortion providers to hold hospital admitting privileges and to meet the standards of ambulatory surgical centers cut the number of facilities providing abortion from 41 in 2012 to 17 in June 2016 (Grossman et al., 2017). The Supreme Court struck down both requirements that month (*Whole Woman's Health v. Hellerstedt*, 2016), but clinics reopened slowly, and one in Austin reopened only in 2017, three years after it closed (NPR, 2017).

Against this unsettled background, Texas became the first state since *Roe* to enforce a ban on abortion at about six weeks of pregnancy (Nash, 2021). Texas Senate Bill 8 (SB8) took effect on September 1, 2021, nearly ten months before *Dobbs*, and banned abortion after about six weeks of gestation. Because private lawsuits rather than state officials enforced the law, no federal court could enjoin an official to stop those suits (*Whole Woman's Health v. Jackson*, 2021*a*). The law nonetheless stood on uncertain legal footing. In declining to block the law as it took effect, the Supreme Court noted "serious questions regarding the constitutionality of the Texas law" and stated that its order was "not based on any conclusion about the constitutionality" of the law (*Whole Woman's Health v. Jackson*, 2021*b*). After *Dobbs*, Texas's ban came to rest on the decision itself, because the state's trigger law, which banned nearly all abortions once *Roe* was overturned, took effect on August 25, 2022 (Supplemental Appendix A1 gives the legal sources).[2]

Then, in June 2022, the Supreme Court held in *Dobbs v. Jackson Women's Health Organization* (2022), a challenge to Mississippi's 15-week ban, that the Constitution confers no right to abortion. Trigger laws and previously enjoined bans then took effect in ten more states (Alabama, Arkansas, Georgia, Idaho, Kentucky, Louisiana, Mississippi, Missouri, South Dakota, and Tennessee) between

[2]Given this background, I treat SB8 as a ban enabled by *Dobbs*, but one enacted earlier.

June and August 2022, joining Oklahoma, which had enforced bans modeled on SB8 since May 2022. West Virginia enacted a total ban in September 2022, and Indiana and South Carolina enforced bans in August 2023 after litigation (Dench, Myers and Pineda-Torres, 2025, Table A.1).

Every ban came with exceptions, but their reach was unclear. Each preserves an exception to prevent the death of the pregnant woman, and some also for a serious risk to her health (Felix, Sobel and Salganicoff, 2024).[3] However, physicians describe the language of these exceptions as too vague to apply at the bedside (Sabbath et al., 2024; Landay et al., 2026), and guessing wrong can mean license revocation, civil liability, or prison for providers (Felix, Sobel and Salganicoff, 2025; Idaho Legislature, 2022). More than 60 percent of obstetrician-gynecologists in ban states report concern about their own legal risk when deciding whether an abortion is medically necessary (Frederiksen et al., 2023), and qualitative work documents clinical decisions that departed from the prior standard of care (Sabbath et al., 2024). However, this uncertainty did not begin with *Dobbs*. Long before the decision, ethics committees at Catholic-owned hospitals denied approval to evacuate the uterus of a miscarrying patient while fetal heart tones were present, forcing physicians to delay care or transfer the patient elsewhere (Freedman, Landy and Steinauer, 2008). In Texas, clinicians interviewed before *Dobbs*, some of them before SB8, described narrow hospital criteria that limited care for patients whose health was at risk (Arey et al., 2023).

B. How Abortion Restrictions Could Affect Maternal Health

Restricting access to abortion, and leaving unclear when exceptions apply, can have wide-reaching effects on maternal health during pregnancy. An early abortion uses either medication, mifepristone followed by misoprostol, or a procedure that empties the uterus, such as uterine aspiration (American College of Obstetricians and Gynecologists and Society of Family Planning, 2020). However, the medications and procedures that make up an abortion are also the standard of care for several conditions unrelated to elective abortion. For example, physicians treat an incomplete, retained, or septic miscarriage with dilation and curettage or with the same medications used for abortion (American College of Obstetricians and Gynecologists, 2018). Before viability, after rupture of the membranes or with preeclampsia with severe features, ending the pregnancy is sometimes the only treatment that prevents infection, sepsis, or the complications of worsening preeclampsia (American College of Obstetricians and Gynecologists and Society for Maternal-Fetal Medicine, 2017; Arey et al., 2023). Additionally, terminations for a lethal fetal anomaly, which is often diagnosed only in the second trimester (Grossman and Chasen, 2020), use the same procedures, dilation and

[3]The bans in Arkansas, Idaho, Mississippi, Oklahoma, South Dakota, and Texas contain no exception for the woman's health, and Texas's 2021 ban allows abortion only for a life-threatening condition.

evacuation or medication induction (Kerns et al., 2012). Some of these pregnancies would also put the woman's health at risk if continued, so these terminations are ambiguously elective.

Further, a pregnancy that does not end in a live birth is common, and its management is routine obstetric care, so many women encounter the procedures a ban restricts. In 2019, an estimated 66.9 percent of U.S. pregnancies ended in a live birth, 13.1 percent in an induced abortion, and 20.0 percent in a miscarriage, ectopic pregnancy, or stillbirth (Rossen et al., 2023). Additionally, even before any formal abortion ban, 12 percent of women obtaining an abortion cited concerns about their own health and 13 percent possible problems with the health of the fetus (Finer et al., 2005). Therefore, restrictions on abortion could affect the care of many pregnant women.

One potential channel through which these restrictions could harm maternal health is delayed management of complications. Because the procedures overlap, a physician unsure whether an exception applies may wait until a miscarriage becomes septic or an ectopic pregnancy ruptures, as in a Georgia death that the state's review committee found preventable (Surana, 2024). Clinicians in states that restrict abortion describe waiting to intervene in ruptured membranes, infection, and severe preeclampsia until the fetus has no cardiac activity or the patient has deteriorated enough to meet a narrow reading of the exception (Sabbath et al., 2024; Arey et al., 2023; Lift Louisiana et al., 2024). Waiting can be costly, and in Texas hospital discharge records, sepsis among women hospitalized with a second-trimester loss rose from 2.9 to 4.5 percent after SB8 (Presser et al., 2025), and reporting describes ectopic pregnancies rupturing before treatment (Seitz, 2024).

A second way restrictions on abortion can harm maternal health is through reduced access to terminations for medical reasons. These can follow the diagnosis of a fetal anomaly or maternal condition, usually later in pregnancy than medication abortion can address. A woman who cannot end such a pregnancy carries it longer, and carrying it is not without risk. In a before-and-after comparison at one Texas center maternal morbidity among patients with a life-limiting fetal condition was 72 percent in 2022 and 2023, when nearly all of them had expectant management, against 60 percent with expectant management and 15 percent with termination in 2017 and 2018 (Nambiar et al., 2025). At three Houston hospitals, sepsis after periviable rupture of membranes rose from 9.4 to 29.2 percent after SB8, again in a before-and-after comparison (Ghafir et al., 2025).

A third channel for harm is the timing of care. Some practices declined to see pregnant patients before 12 weeks to avoid managing early losses (Lift Louisiana et al., 2024). If practices stop seeing women early in pregnancy, care could deteriorate for every pregnancy. Additionally, obstetrician-gynecologists could leave or stop seeing pregnant patients. However, counts of obstetrician-gynecologists rose about as much in ban states as in protected states after *Dobbs* in a

descriptive comparison (Staiger et al., 2025), although one difference-in-differences estimate finds a relative decline in the most restrictive states (Zhu et al., 2025).

II. Data

A. Insurance Claims

To study the effects of the bans on maternal health, care, and pregnancy outcomes, I use Inovalon closed claims real-world data, which are comprised of post-adjudication medical and pharmacy claims received directly from more than 160 primary payer sources, including national, regional, and local health plans across all 50 states, the District of Columbia, and Puerto Rico. The dataset includes more than 246 million covered lives over a 10-year lookback across commercial, Medicare Advantage, and Medicaid lines of business. Patient identity is resolved before de-identification using a conservative deterministic matching approach, producing a persistent identifier that follows each individual across health plans and over time. Only birth year, gender, and three-digit ZIP code carry into the de-identified dataset. Data are de-identified under the HIPAA Expert Determination standard (45 CFR §164.514(b)(1)). Available elements include claim type, place of service, service and admission and discharge dates, diagnosis and procedure codes, DRG, HCPCS, NDC, proxy cost data, enrollment and continuous enrollment, insurance type, ACA indicators, and a provider's National Provider Identifier (NPI), taxonomy, specialty, and facility type (Inovalon, 2026).

Within these data, the claims file records each woman's health care, and the enrollment file records her demographics, state of residence, and payer each month. I also merge in the National Plan and Provider Enumeration System (NPPES), the federal NPI registry, for each provider's practice address by year (Centers for Medicare & Medicaid Services, 2026).

These data provide unusually rich detail on the health care of millions of insured women. In my analysis sample, I record the services billed to a woman's insurance from a year before conception through 42 days after her pregnancy ends. Unlike birth certificates, which record only pregnancies that end in a birth, these claims also record pregnancies that end in a loss or an insurance-billed abortion, along with every diagnosis a woman received in the year before her pregnancy began. However, one feature of these data is that they are a convenience sample, meaning that insurers contribute data under contracts with the data provider that begin and end over time. Because the data carry no plan identifier, I cannot distinguish a woman who leaves her plan from one whose plan stopped contracting with the data provider. Additionally, because insurance plans can be regional, such as Medicaid managed care plans that each state contracts for separately (Medicaid and CHIP Payment and Access Commission, 2019), an insurer's entry or exit can add or remove a whole

population and potentially shift the risk profile of a state's members. I address this by restricting my analysis sample to women continuously enrolled from 2020 through 2024 (Section II.D).

I construct two analysis samples: the main analysis sample of pregnancies, with one observation per pregnancy episode of a continuously enrolled woman, and a panel of continuously enrolled women aged 15 to 45 by state and quarter for pregnancy and birth rates (Section VI.A).

B. Ban and Protected States

I group thirteen treated states into three cohorts: Texas; the eleven states that banned abortion directly after *Dobbs*; and West Virginia.[4] I compare them with 24 states and the District of Columbia where abortion remained legal through at least 22 weeks of gestation throughout the period (Curhan, 2026). I exclude from my analysis eleven states whose bans were enjoined, partial, or changing during the window. The protected states enter only as never-treated controls. Supplemental Appendix Table A2 further describes the rationale behind the choice of treatment, control, and excluded states, and Table A1 lists each state with its effective date and statute.

C. Constructing Pregnancy Episodes

I build pregnancy episodes following published claims-based algorithms (Moll et al., 2021; Ailes et al., 2023; Makadia et al., 2026). I begin by using pregnancy-specific codes to identify each pregnancy and date its start and its ending.[5] I then gather the woman's complete claim history around these dates. Each episode has three windows. The pre-pregnancy window covers the 365 days before the pregnancy start and supplies the measures of health before pregnancy, the pregnancy window runs from the start to the ending, and the postpartum window runs from the day after the ending through 42 days. This process captures all of the insured care a woman receives during pregnancy, not just pregnancy-coded care, since providers bill some care without a pregnancy code. For example, an emergency department may bill sepsis with only the presenting condition, so pregnancy-coded claims alone would miss it.

Endings

I identify each ending from diagnosis, procedure, and diagnosis-related-group codes for a live birth, stillbirth, molar pregnancy, ectopic pregnancy, induced abortion, or spontaneous abortion,

[4]Because Indiana and South Carolina banned abortion in August 2023 and the data end in December 2024, I cannot follow a complete post-ban window of their pregnancies to its ending, so I also exclude them from the pregnancy estimates.

[5]The pregnancy and ending codes draw on the Maternal and Infant Health code lists that the Centers for Medicare & Medicaid Services publish for identifying pregnant and postpartum beneficiaries (Centers for Medicare & Medicaid Services, 2023), with the ending hierarchy of Moll et al. (2021). Supplemental Appendix Table A21 lists the codes by outcome.

and where an ending carries codes for more than one type, I apply the hierarchy that Moll et al. (2021) validated against physician-adjudicated records. Because the abortion codes include the billing codes for mifepristone and misoprostol, which also treat miscarriage (American College of Obstetricians and Gynecologists, 2018), the classification errs toward counting too many abortions rather than too few. The order of the hierarchy moves the share of pregnancies classified as miscarriages by a few percentage points, while induced abortions stay below 5 percent of commercially insured pregnancies under every order (Fuerst, Schrote and Rodriguez, 2026). Supplemental Appendix A2 describes how I link ending codes and exclude endings that imply a pregnancy too close to the previous one.

Pregnancy start

I date each pregnancy from the first day of the estimated last menstrual period (American College of Obstetricians and Gynecologists, 2017), and call this date the pregnancy start. I date most pregnancies from Z3A diagnosis codes, which record the gestational week at a visit, using a validated algorithm with a median absolute error of four days (Chomistek et al., 2023). Dating is least accurate for early losses. Supplemental Appendix A2 gives the full procedure and the share of pregnancies dated by each source.

Pregnancies without an observed ending

Some pregnancies carry pregnancy care but no ending code. In my core analysis sample, they are 6.2 percent of ban-state pregnancies conceived in the six quarters before each state's ban (Section II.D). I build these episodes from clusters of pregnancy-indicating claims, such as prenatal visits and gestational-age codes, that fall outside every other observed pregnancy for a woman, as Sarayani et al. (2020) do for pregnancies with unknown outcomes, and end each on the date of its last such claim (Supplemental Appendix A2).

I keep pregnancies without an observed ending rather than excluding them, because their share is itself informative. For a woman who is insured throughout, a pregnancy with prenatal care and no billed ending is most plausibly one that ended in care that insurance did not pay for, such as an abortion paid out of pocket, obtained out of state, or obtained by mail, or a loss that received no care. I exclude them from the outcomes that require an observed ending, namely the live-birth, miscarriage, and insurance-billed abortion shares, preterm birth, the length of pregnancy, and the days to the first visit.

D. Analysis Samples

In the core analysis sample, the *continuous-enrollment sample*, I keep only the pregnancies of women continuously enrolled from January 2020 through December 2024. This requirement lets me observe, for nearly every pregnancy, the full year preceding it and the full postpartum period.[6] Holding the same women in the data throughout also keeps the sample's membership from shifting with other changes in who appears in the claims over the years the bans took effect.

While the continuous-enrollment sample keeps a constant group of women at risk of pregnancy, each observed in detail, it is not without trade-offs. First, it retains only 19 percent of the pregnancies that pass every other restriction when I impose no enrollment requirement (995,377 of 5,325,806). Second, it may select certain kinds of women. For example, women with stable coverage, whether through Medicaid or a commercial plan, are plausibly better able than most to travel around a ban, and nationally, continuous coverage is more common among higher-income and married women (Daw et al., 2017). Despite this, the pregnancies in my sample broadly resemble all U.S. births on a number of demographic features (Supplemental Appendix Table A7). For example, 22.7 percent of sample pregnancies are to women under 25, against 21.1 percent of U.S. births, and 15.1 percent are to Black women, against 13.7 percent (Osterman et al., 2025). The sample also includes more Medicaid-covered pregnancies than the nation does, as Medicaid covers 50.7 percent of sample pregnancies against 41.5 percent of U.S. births (Osterman et al., 2025). If the bans harmed Medicaid-covered women more, this would push the estimates toward larger effects rather than smaller ones.

In addition, my enrollment requirement excludes, by construction, every woman who died before December 2024, because a woman who dies loses her enrollment.[7] My sample therefore lacks pregnancies that ended in or led to a maternal death, the most severe outcomes, and so tilts slightly toward healthier pregnancies. If the bans raised maternal deaths, the excluded deaths would come disproportionately from ban states after the bans, and the estimates for severe maternal morbidity would understate the true effect. However, maternal death, at 17.9 per 100,000 live births in 2024 (Hoyert, 2026), is more than a hundred times rarer than severe maternal morbidity, at 2.6 percent of ban-state pregnancies in my sample before the bans, so the understatement is small.

[6] Supplemental Appendix A3 reports how often I fully observe the pre-pregnancy year for pregnancies that start in 2020.
[7] Additionally, the data do not let me identify deaths directly, because the data provider withholds them for privacy.

Additional restrictions

I keep pregnancies that started from January 2020 through February 2024 in the 15 ban states (the 13 treated states, Indiana, and South Carolina) and 25 protected jurisdictions, to follow a typical 40-week pregnancy to its ending, and I exclude observed endings with an implied gestation shorter than 28 or longer than 315 days, likely coding or dating errors.[8] The continuous-enrollment sample contains 995,377 pregnancies, and Supplemental Appendix A3 reports counts by step.

Demographics, health, and treatment

A woman's demographics for each pregnancy come from the enrollment file in the month the pregnancy started. I control for age and its square;[9] race and ethnicity, in six mutually exclusive groups (Hispanic of any race, and non-Hispanic White, Black, Asian, other or multiple races, and missing);[10] Medicaid coverage; indicators for a first observed pregnancy and a multiple gestation; and a pre-pregnancy comorbidity score. I measure this score from available claims in the 365 days before the pregnancy start, using the conditions of the obstetric comorbidity index of Leonard et al. (2020), a validated predictor of severe maternal morbidity. Supplemental Appendix A3 lists the conditions and their weights.[11]

I assign treatment by the state of residence in the enrollment file rather than by where care was delivered, because claims locate care only through the provider's registered NPI address (Supplemental Appendix A3). I use the state recorded in the month the pregnancy started. Although the enrollment file records a state each month, it never changes for a woman in these data and records her most recent state, so a woman who moved from a ban state to a protected state would count as untreated, biasing the estimates toward zero. To address this, in the close-to-care design (Supplemental Appendix A9) I assign location from the providers a woman saw before conception.

Several of these demographic and health characteristics define the subgroups in Section V.D. Medicaid-covered pregnancies are those with a Medicaid payer in the month the pregnancy started, minority women are those with a Black race code or a Hispanic ethnicity code, and high-risk women are those with at least one condition in the obstetric comorbidity index, about 42 percent of ban-state pregnancies before the bans.

[8] These exclusions remove about 0.1 percent of the remaining pregnancies.

[9] Age enters as a quadratic because the risk of severe maternal morbidity rises nonlinearly with maternal age, low from the early twenties to the mid-thirties and rising steeply after about 39 (Lisonkova et al., 2017).

[10] The data provider imputes race and ethnicity using a proprietary algorithm, and they can be unknown.

[11] One limitation of this measure is that a condition appears in the claims only if the woman sought care for it, so women who rarely see a doctor may appear healthier than they are. However, nearly every woman in my sample is insured throughout the year before conception, and its authors validated it on hospital records with the same limitation.

E. Outcomes

Each channel in Section I.B predicts a pattern of maternal harm from abortion restrictions that may be observable in health care claims. I therefore define twelve outcomes, in three groups, for the parts of pregnancy that abortion bans are most likely to affect. Supplemental Appendix Table A21 provides code-level definitions and sources for each.

Maternal health

Severe maternal morbidity measures the life-threatening complications at the center of the warnings, such as sepsis and hemorrhage, and would rise if delayed management let a miscarriage become septic or an ectopic pregnancy rupture. To define it, I use the Centers for Disease Control and Prevention's definition (Centers for Disease Control and Prevention, 2024), with the ICD-10 codes for its diagnosis indicators listed in Leonard et al. (2020, Appendix 1), and count a pregnancy as having severe maternal morbidity if any of its 16 diagnosis indicators, among them sepsis, shock, eclampsia, and acute renal failure, appears on any claim from the pregnancy start through 42 days after the ending.[12] Because severe morbidity is rare, affecting about 2.6 percent of ban-state pregnancies before the bans, I also measure a composite of more common *adverse pregnancy outcomes*, adapted from Crump, Sundquist and Sundquist (2024). It captures the harms of a pregnancy carried longer or managed later. If terminations for medical reasons became harder to obtain, pregnancies that would have ended in a managed termination would instead end in a complicated birth or loss, and a delayed miscarriage can become infected. It equals one for a diagnosis of infection or sepsis, preterm birth, gestational diabetes, poor fetal growth, or placental abruption.

I also study two of the most common serious complications of pregnancy separately. *Hypertensive disorders of pregnancy* are the most common, affecting 21 percent of ban-state pregnancies before the bans. They are disorders of high blood pressure that arise in pregnancy, among them gestational hypertension and preeclampsia, and in their severe forms they can damage the kidneys, liver, and brain. Delivery or termination is the treatment for these severe forms, so delayed management could let them progress. They are also the outcome earlier restrictions on abortion providers affected (Gardner, 2024), so they give the sharpest comparison with prior work.

Finally, I measure *preterm birth* only among live births. Because the bans may themselves change the mix of pregnancy endings, this outcome isolates harm to pregnancies carried to delivery. Lost access to terminations for medical reasons may show up here, as a pregnancy with a severe anomaly or complication that would have been ended instead continues until an early delivery.

[12] Supplemental Appendix A10 explains why I omit the definition's five procedure indicators.

Timing and intensity of care

The second group of outcomes asks whether the timing and intensity of care changed. Delayed management or deferred early care would appear here first, before or without a change in health. The *days to the first visit*, the log number of days from the pregnancy start to the woman's first claim of any kind during the pregnancy, tests whether practices deferred early care, as some practices in ban states said they would to avoid managing early losses (Lift Louisiana et al., 2024).[13] A *prolonged delivery stay*, a live birth whose delivery admission ran longer than two days after a vaginal delivery or four after a cesarean, the 48 and 96 hours that federal law requires private health plans to cover (U.S. Code, 1996), tests for delayed management of complications. Such delays would lengthen hospital stays at delivery. The log *length of pregnancy* and the log *number of visits* describe how long a pregnancy lasted and how much care it received. However, I expect these two to rise mechanically when fewer pregnancies end early, so I read them alongside the endings rather than as measures of the quality of care.[14]

Pregnancy endings

The third group of outcomes records how each pregnancy ended. A ban that prevented terminations should change how pregnancies end, with fewer insurance-billed abortions and more pregnancies ending in a live birth or a miscarriage or with no observed ending.[15] I study four outcomes. The first three record whether a pregnancy ended in a *live birth*, a *miscarriage*, or an *insurance-billed abortion*, and the fourth whether its *ending was not observed* (Section II.C). I identify an insurance-billed abortion from any diagnosis, procedure, or medication code for an induced abortion. The first three are shares of pregnancies with an observed ending and the fourth is a share of all pregnancies, so the four do not sum to one. Insurance-billed abortion is the most direct measure of the policy for insured women, although it captures only a fraction of all abortions (Roberts et al., 2014; Abernathy, Rodriguez and Swartz, 2025). As Section II.C explains, this measure errs toward counting too many abortions rather than too few.

[13] Payers commonly reimburse prenatal care and childbirth at a global or bundled rate (CPT 59400, 59510, 59610, 59618), so providers often do not bill each prenatal visit separately (Gourevitch et al., 2020). Prenatal visits under such a package may appear only as zero-charge tracking claims or not at all, so the number of visits and the days to the first visit may be undercounted for these pregnancies. This undercount would bias the estimates only if the use of global billing changed differently in ban and protected states after the bans.

[14] I log these outcomes because their distributions have a long right tail, and in levels a few extreme pregnancies would drive the estimates. I additionally winsorize counts at the 99th percentile.

[15] Ectopic, stillbirth, and molar endings, 2.5, 0.9, and 0.1 percent of observed endings, are too rare to estimate.

III. Descriptive Patterns

Table 1 reports means in the continuous-enrollment sample for ban and protected states before and after the bans; the protected-state post-period begins with pregnancies that started in or after June 2022. Before the bans, ban-state women were younger than protected-state women (29.0 against 30.4 years), and the racial and ethnic composition of insured pregnancies differed between the two groups of states. Medicaid covered 42.4 percent of ban-state pregnancies before the bans and 53.4 percent in protected states, and ban-state women entered pregnancy with a higher mean comorbidity score, 8.7 against 8.4. Supplemental Appendix Table A6 reports normalized differences (Imbens and Wooldridge, 2009) for these characteristics in the pre-period.[16]

In addition to these baseline differences in characteristics, the pregnancies themselves differed, and the gaps persist in the rows of Table 1 adjusted for characteristics. First, the difference in how pregnancies ended is striking. Before the bans, insurance-billed abortion ended about 1 percent of ban-state and 14 percent of protected-state pregnancies with an observed ending in 2020, and a live birth ended a correspondingly larger share of ban-state pregnancies. Some of this gap likely reflects ban-state pregnancies that ended in an abortion paid outside insurance, and some likely reflects differences in the insured populations the data cover in each state.

Second, ban-state pregnancies fared worse, particularly in the states that banned at *Dobbs* rather than in Texas (Figure 1). They more often involved an adverse pregnancy outcome and a hypertensive disorder. Part of this gap perhaps follows from more ban-state pregnancies continuing to term, and so facing the complications of later pregnancy, but even among live births, and adjusted for characteristics, preterm birth was more common in ban states before the bans. Finally, ban-state pregnancies lasted longer, with about as many visits and prolonged delivery stays, and the first visit came at about the same point in both groups of states in the raw means but later in ban states once adjusted for characteristics. Even with these adjustments, ban-state pregnancies before the bans more often had an adverse pregnancy outcome or a hypertensive disorder and, among live births, a preterm birth, while severe maternal morbidity was similar in the two groups of states.

Figure 1 shows the raw series for the twelve outcomes over calendar time for Texas, the states that banned at *Dobbs*, and protected states. The gaps were largely stable, and in some cases widening, before the bans. For maternal health and for the timing and intensity of care, the ban-state and protected-state series move roughly in parallel, apart from the length of pregnancy, the number of visits, and prolonged delivery stays. However, there are clear divergences in the

[16]These differences reflect both the underlying demographic mix of each group of states and differences in the insured populations the data cover. The adjusted rows of Table 1 hold these characteristics fixed.

pregnancy endings. The ban-state series for insurance-billed abortion and live birth are nearly flat, while in protected states the live-birth share fell and insurance-billed abortion rose, as several states widened protections and coverage (Supplemental Appendix A1) and demand for abortion by telehealth rose during the COVID-19 pandemic (Aiken et al., 2020). No maternal health or care series shows a sharp break at either ban, and Section V tests whether the smaller movements in the adverse-outcome composite and preterm birth exceed the changes in protected states.

IV. Empirical Strategy

I compare pregnancies in ban states with pregnancies in protected states, before and after each ban. The unit of observation is a pregnancy p in state s conceived in quarter q. Because the bans took effect at different times, I estimate the stacked difference-in-differences design of Wing, Freedman and Hollingsworth (2024).[17] For each adoption quarter a, the first fully treated quarter of one or more ban states (the fourth quarter of 2021 for Texas, the third quarter of 2022 for the *Dobbs* cohort, and the fourth quarter of 2022 for West Virginia),[18] I build a sub-experiment containing those states and, as controls, the protected states. A ban state never serves as a control. Event time is $e_{pa} = q - a$, with a window of $e_{pa} \in [-6, 4]$. A sub-experiment enters the stack only if its full window lies inside the data and it has at least one clean control, so I estimate every event-time coefficient on the same cohorts. The 818,245 pregnancies conceived inside these windows form the *event-study sample*. Supplemental Appendix A8 gives each cohort's window, and Supplemental Appendix Table A14 the pregnancies in each sub-experiment. I then estimate

$$Y_{psqa} \;=\; \sum_{h=-6}^{4} \delta_h \cdot \mathbf{1}\{e_{pa} = h\} \cdot D_{sa} \;+\; \sum_{h} \lambda_h \cdot \mathbf{1}\{e_{pa} = h\} \;+\; X_p'\gamma \;+\; \phi_{sa} \;+\; \varepsilon_{psqa},$$

where Y_{psqa} is an outcome of pregnancy p, conceived in quarter q by a woman living in state s, in sub-experiment a, and h indexes event time. D_{sa} indicates the treated states of sub-experiment a, λ_h are event-time effects shared by treated and control states, ϕ_{sa} are sub-experiment by state fixed effects, and X_p contains the individual controls described in Section II.D. The sub-experiment by state fixed effects hold fixed each state's average outcome within each sub-experiment, so the estimates come from changes within a state over its own window rather than from the gaps between

[17] Stacking compares each treated cohort only with never-treated states and avoids the comparisons with already-treated states that bias two-way fixed effects when treatment is staggered and effects vary over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). Supplemental Appendix Figure A4 compares the two.

[18] The third quarter of 2022 is only partly treated for Georgia, Idaho, and Mississippi, whose bans took effect in July and August, and for Kentucky and Louisiana, where state courts blocked enforcement for much of July, so the estimate at $h = 0$ may be attenuated for these five states. Oklahoma, by contrast, enforced bans from May 2022, so its quarter -1 is already treated, one reason I compare with the average of all six pre-ban quarters and, in Supplemental Appendix A8, with quarter -2. The later event-study coefficients, in fully treated quarters, show whether an effect appears once the bans were fully in force.

ban and protected states in Section III. The event-time effects, common to every sub-experiment, absorb changes that ban and protected states share, so δ_h measures how far ban-state pregnancies departed from that shared path at event time h.

Rather than omitting quarter -1, I normalize the coefficients so that the six pre-ban coefficients average zero, $\sum_{h=-6}^{-1} \delta_h = 0$, so I measure each coefficient, δ_{-1} included, against the pre-ban average, with its own standard error. Averaging over several reference quarters also reduces the noise that a single reference quarter adds to every coefficient (Miller, 2023). The joint pre-trend test, that δ_{-6} through δ_{-1} are all zero, is the same under either normalization. When it fails, I read the failure partly as treatment reaching the last pre-ban quarters, through anticipation and through errors in dating pregnancies, rather than only as diverging trends, so I also look at whether the pre-period in each event study is flat or trending. I weight control observations in sub-experiment a by $(N_a^D/N^D)/(N_a^C/N^C)$, the sub-experiment's share of treated observations divided by its share of control observations. Without these weights a stacked regression does not identify a convex combination of cohort-specific effects, and with them, under parallel trends and no anticipation, δ_h is the trimmed aggregate average treatment effect on the treated at event time h (Wing, Freedman and Hollingsworth, 2024). The weights give each sub-experiment's controls the same share of the weighted control group that its ban states have of the treated group, and the sample sizes I report count each weighted control observation. For the pooled estimate I replace all the event-time interactions with one indicator $D_{sa} \cdot \mathbf{1}\{e_{pa} \geq 0\}$, on the same stacked sample and with the same weights, so the pooled and event-study estimates use the same pregnancies.

I compare the post-ban average with the pre-ban average, rather than with the last pre-ban quarter as in Wing, Freedman and Hollingsworth (2024), because quarter -1 is likely partly treated. The main reason is that a pregnancy carried to term lasts about 40 weeks, so a pregnancy conceived in quarter -1 or -2 is still under way when the ban takes effect and those quarters are partly exposed to it. Physicians and patients may also have acted in anticipation of a ban (Section I.A), and because pregnancy start dates in claims are noisy, most of all for early losses, some pregnancies assigned to quarter -1 may have been conceived after the ban (Supplemental Appendix A2). Averaging over all six pre-ban quarters dilutes these partly exposed quarters rather than letting one of them set the reference. Because the partly exposed quarters remain in the average, any effect they carry pulls the estimates, and their upper bounds, toward zero. Supplemental Appendix Table A15 reports estimates against quarter -2, which is less exposed to the bans than quarter -1. I cluster standard errors by state rather than by sub-experiment and state, as the same control

states appear in several sub-experiments.[19] I report two specifications. The Raw specification omits X_p. The Controls specification is the equation above, and it is my preferred specification because it measures the effect of a ban among observably similar women.

A causal reading of these estimates requires that, absent the bans, outcomes in each treated cohort would have followed the path of its controls. Several features of the setting strain this assumption. The first is that for some outcomes the trends clearly do not move together before the bans. This is most true for insurance-billed abortion, whose protected-state series rises well before the bans (Section III), consistent with the argument that it was already *Dobbs* in ban states, so I read its estimate as describing ongoing patterns. The live-birth and miscarriage shares also move before the bans, though less, so I read their estimates as suggestive. The length of pregnancy and the number of visits, both rising with the share of pregnancies carried to term, the share of prolonged delivery stays, and the share with no observed ending do not, so I read their estimates with the same caution. However, the maternal health outcomes and the days to the first visit move together before the bans, so I read their estimates around the ban dates as effects of the bans.

A further strain falls on the stable unit treatment value assumption, under which the bans must not change outcomes in the comparison states. However, protected states absorbed travel from ban states and expanded access to abortion after *Dobbs*. Expanded access would bias the estimates toward finding harm, and crowding of protected-state providers by ban-state patients would bias them toward zero. To address this, in the close-to-care design of Supplemental Appendix A9 I compare women within ban states. Additionally, in case protected states compare poorly, in Supplemental Appendix A8 I reweight protected states and pre-ban quarters to match ban states' pre-ban paths with the synthetic difference-in-differences of Arkhangelsky et al. (2021).

V. The Effects of the Bans

The bans changed maternal health and the timing and intensity of care little, while modestly shifting pregnancies toward a live birth. I report pooled estimates from the Controls specification in the continuous-enrollment sample, each scaled by the pre-ban mean, with the quarterly event studies (Figure 2). Table 2 also reports the Raw specification and the p-value of a joint test that the pre-period coefficients are zero.

[19] With 38 state clusters, 13 of them treated and two of the three cohorts a single state, cluster-robust standard errors can understate uncertainty (Conley and Taber, 2011; MacKinnon and Webb, 2018; Ferman and Pinto, 2019), so I treat p-values near conventional thresholds with caution.

A. Pregnancy Endings

If the bans prevented terminations, the most direct place to see this is abortion billed to insurance, although it is an imperfect proxy. However, insurance-billed abortion was already so rare in ban states that it had little room to fall. The pooled estimate is 2.5 percentage points lower ($p < 0.001$), or -237.1 percent of the pre-ban mean of 1.06 percent, a magnitude larger than the mean that is possible only because I measure the estimate against protected states. The raw series show that the protected-state side accounts for most of this (Figure 1). In ban states, an insurance-billed abortion ended between 0.8 and 1.2 percent of pregnancies with an observed ending in every quarter from 2020 through 2023, while in protected states the share rose from 14 percent in 2020 to 17 percent in the quarter of the decision and 20 percent by the end of 2023. The event study traces the same divergence and rejects parallel pre-trends ($p < 0.001$), so I read this estimate as describing ongoing patterns rather than an effect of the bans.

Additionally, if the bans moved terminations outside the insurance system, to other states or to medication by mail, those pregnancies may appear as a higher share of pregnancies with care but no observed ending, because in the continuous-enrollment sample every woman is insured and enrolled throughout. In Texas this share was 4.3 percent among pregnancies conceived from 2020 through the first quarter of 2021, rose to 8.4 percent among pregnancies conceived in the three quarters around SB8, and returned to 4.4 percent afterward (Figure 1). Over the same quarters it moved from 4.7 to 5.2 percent in protected states and from 6.2 to 7.0 percent in the states that banned at *Dobbs*, so states whose laws had not yet changed share part of the Texas rise. No comparable rise follows *Dobbs*. One possibility is that by mid-2022 women who meant to end a pregnancy could do so before any prenatal care was billed, or after a single visit that was not enough to form an episode, and increasingly through telehealth (Society of Family Planning, 2026). Others ended pregnancies outside the formal health care system altogether, as provision of medications for self-managed abortion rose by about 28,000 above its expected level in the six months after the decision, in a descriptive study with no comparison group (Aiken et al., 2024). The pooled estimate for this share is -13.3 percent ($p = 0.08$), and its 95 percent confidence interval rules out increases larger than 1.6 percent, although its pre-period coefficients jointly differ from zero ($p = 0.001$).

Lastly, the share of pregnancies with an observed ending that ended in a live birth rises 2.7 percent ($p < 0.001$; 95 percent confidence interval 1.7 to 3.8 percent). The miscarriage share would rise if pregnancies that would have been terminated instead ended in a loss, and it changes by 3.0 percent, with an interval from -0.9 to 6.9 percent. The live-birth estimate is weaker than its p-value suggests. Its event study sits 1 percent of the pre-ban mean below the pre-ban average

from six to four quarters before the bans and 1 percent above it from three quarters before (joint $p < 0.001$), and without Texas the estimate falls to 2.3 percent ($p = 0.005$). The rise also does not appear in the episode-enrolled sample, where women moving in and out of the data change the sample (Supplemental Appendix A4), and synthetic difference-in-differences gives a smaller, less precise estimate whose interval still contains the main one (Supplemental Appendix A8). I therefore read the rise in live births as suggestive. This share describes how pregnancies ended, not how many women gave birth, so Section VI.A turns to birth and pregnancy rates.

B. Maternal Health

The only suggestion of harm in the maternal health estimates is in the composite of common adverse pregnancy outcomes. The estimate is 0.008, or 2.8 percent of the pre-ban mean, significant at the 5 percent level ($p = 0.01$), and 3.2 percent ($p = 0.007$) in the Raw specification. Its event study is flat before the bans and positive in every quarter after (Figure 2), and the 95 percent confidence interval rules out increases larger than 4.9 percent. Most of this rise is accounted for by the shift in endings rather than by worse outcomes for each pregnancy. Before the bans, 36.9 percent of live births had one of these outcomes, against 2.2 percent of losses, so among pregnancies with an observed ending the 2.2-percentage-point rise in the live-birth share would by itself raise the composite by about 0.75 percentage points at pre-ban rates, most of the 0.79-point estimate. Within live births, the composite changes by −1.2 percent, with an interval that rules out increases larger than 0.6 percent (Supplemental Appendix Table A10). None of its five components rises significantly on its own. Poor fetal growth, a diagnosis made only in pregnancies that continue, rises 7.4 percent ($p = 0.12$), while infection or sepsis, the complication delayed management would produce, falls 6.8 percent ($p = 0.19$; 95 percent confidence interval −17.0 to 3.4 percent; Supplemental Appendix Table A11), although its pre-period coefficients jointly differ from zero ($p = 0.03$).

Severe maternal morbidity, the life-threatening outcome at the center of the warnings, changes by −7.6 per 10,000 pregnancies ($p = 0.49$), or −2.9 percent of a pre-ban rate of 264.5 per 10,000, and the 95 percent confidence interval rules out increases larger than 5.6 percent. Hypertensive disorders of pregnancy, the most common serious complication and the outcome earlier restrictions affected, likewise change little. The estimate is 1.0 percent ($p = 0.42$), and the 95 percent confidence interval rules out increases larger than 3.4 percent, below the 11 to 16 percent rise that earlier restrictions on abortion providers produced (Gardner, 2024). Hypertensive disorders rose in both groups of states over the period (Figure 1), and the bans added little to that trend in ban states relative to protected states. Preterm birth among live births changes by −0.9 percent, and its interval,

from −4.5 to 2.6 percent, likewise rules out large increases in a common adverse outcome. Because preterm birth and prolonged delivery stays are measured only among live births, which the bans also changed, I read these two estimates as suggestive.

C. Timing and Intensity of Care

The raw series for the timing and intensity of care show no clear break at either ban (Figure 1), and where the gaps with protected states widen, they widen gradually, mostly because the protected-state series fall. The days to the first visit would lengthen if practices deferred early care. They change by 1.4 percent ($p = 0.27$), with pre-period coefficients that do not jointly differ from zero (joint $p = 0.12$), and the interval rules out delays longer than 3.8 percent.

Hospital stays at delivery, if anything, shortened. The share of live births whose delivery stay ran past this covered stay falls 4.6 percent ($p = 0.03$; 95 percent confidence interval −8.6 to −0.6 percent), but its pre-period coefficients jointly differ from zero ($p = 0.03$), so I do not read the decline as an effect of the bans. Whether or not the decline reflects the bans, the estimate gives no sign that physicians managed complications later.

The length of pregnancy rises 3.0 percent ($p < 0.001$), as expected when fewer pregnancies end early, and the number of visits is 3.3 percent higher ($p = 0.03$; 95 percent confidence interval 0.4 to 6.2 percent). Both rise before the bans as live births do (Section V.A; joint $p < 0.001$ for the length of pregnancy and < 0.001 for visits), so, as anticipated in Section II.E, I read them as part of the shift in endings rather than as evidence about the quality of care. None of these estimates points to a change in how physicians managed pregnancies on average.

D. Heterogeneity Analysis

A small average effect can hide large effects in particular groups, so I define five subgroups whose effects have reason to differ from the overall average. I first look at Medicaid-covered pregnancies, because women with low incomes may be the least able to travel for care or to pay for an abortion out of pocket (Jerman, Jones and Onda, 2016). I next look at minority women, Black and Hispanic women reported together, whose outcomes were worse before the bans. Black women in the United States die of maternal causes at about three times the rate of White women (Hoyert, 2026) and experience severe maternal morbidity at nearly twice the rate (Leonard et al., 2019). I then look directly at women who enter pregnancy with a high-risk condition, who are the most likely to need potentially lifesaving care, such as a termination for medical reasons, that the bans may have called into question. The opposite concern is that the bans led many young, healthy women to

carry pregnancies to term. Such births could mask worse outcomes for everyone else, so I look at pregnancies to women under 25. Finally, I exclude Texas, whose ban came first, was enforced through private lawsuits, and was in place longest, to check whether the results hold without it.

For each group, I rerun the stacked design and report the percent change over the group's own pre-ban mean, so each estimate is the change for that group rather than its contribution to the pooled effect (Figure 3 and Supplemental Appendix Table A9). The estimates are similar across these groups. For each of the twelve pregnancy outcomes, every subgroup's 95 percent confidence interval contains the pooled point estimate, apart from the under-25 interval for the adverse-outcome composite, which lies just above the pooled estimate (lower end 2.8 percent). Without Texas the estimates are less precise but point the same way, as the composite rises 3.4 percent with an interval from −0.5 to 7.3 percent, and the interval for severe maternal morbidity reaches 11.8 percent (Supplemental Appendix Table A9). The composite rises 4.2 percent among Medicaid pregnancies ($p = 0.02$), 4.0 percent among high-risk women ($p = 0.04$), and 6.3 percent among women under 25 ($p < 0.001$), the only subgroup estimates for maternal health whose intervals exclude zero. Severe maternal morbidity among high-risk women, the group of most concern, changes by −2.2 percent ($p = 0.64$; 95 percent confidence interval −11.4 to 7.1 percent).

Because the bans changed how pregnancies ended, I also estimate the health and care outcomes within live births and within losses (miscarriage, stillbirth, and ectopic pregnancy; Supplemental Appendix Table A10). These estimates condition on an ending the bans changed, so they help interpret the pooled estimates rather than isolate effects of their own. Within live births, the length of pregnancy changes by −0.1 percent, and the number of visits by −2.0 percent, so the pooled rises in these outcomes mostly reflect more pregnancies carried to delivery. Losses, the small group the warnings most concerned, yield imprecise estimates. Among them, severe maternal morbidity changes by 11.3 percent, with an interval that does not rule out a rise of 33.4 percent, and the first visit comes 2.6 percent later ($p = 0.05$). These intervals cannot rule out that more losses ended in a severe complication after the bans.

VI. The Fertility Channel

Beyond the outcomes in Section V, the bans could also change how many women carried pregnancies to term and who became pregnant.

A. *Fertility*

A ban that prevented abortions, insured or not, should raise births, as earlier restrictions on abortion access do (Fischer, Royer and White, 2018; Venator and Fletcher, 2021). Conceptions may also fall if women use more contraception after a ban (Xu et al., 2024; Gallen and Lu, 2025), although in a difference-in-differences comparison, fills of oral contraceptives fell 4.1 percent more in states with total bans (Qato et al., 2024), so the net effect on births is ambiguous. I measure the net effect by estimating the stacked design on the panel of enrolled women aged 15 to 45 by state and quarter, weighted by enrollment. The outcome is live births or pregnancies per 1,000 enrolled women per year, counted by the month the pregnancy started, and I report each estimate in percent of its pre-ban mean (Supplemental Appendix Table A12). The birth rate is 3.7 percent higher relative to protected states ($p = 0.02$; 95 percent confidence interval 0.7 to 6.7 percent), the same sign as the 2.3 percent increase that Dench, Pineda-Torres and Myers (2024) estimate from birth certificates. I read this rise as suggestive, because its pre-ban coefficients jointly differ from zero ($p < 0.001$) and synthetic difference-in-differences, which has less power, gives 6.2 percent with an interval from -2.8 to 15.2 percent (Supplemental Appendix Table A17). The raw rates in panels (c) and (d) of Figure 4 show where this estimate comes from. Both the pregnancy rate and the birth rate fall over the period in both groups of states, as the women in a continuously enrolled panel age, and birth rates in ban and protected states move together through mid-2022. From late 2022 on they separate, as the ban-state birth rate holds near its 2022 level while the protected-state rate keeps falling, so relative to protected states, births held up in ban states.

The pregnancy rate counts conceptions, not births. It changes by 0.2 percent ($p = 0.86$; 95 percent confidence interval -2.2 to 2.6 percent). Comparing the two rates shows where any added births came from. Together with the rise in the live-birth share of observed endings (Section V.A), these estimates suggest that about as many women became pregnant after the bans but more of their pregnancies ended in a live birth. Across the subgroups in Supplemental Appendix Table A12, the birth-rate estimate is always positive. Among women with a high-risk condition, the birth rate rises 5.5 percent ($p = 0.002$) but their pregnancy rate only 1.6 percent ($p = 0.21$), so they appear to have carried more of their pregnancies to term rather than conceived more often.

B. *Who Became Pregnant*

The bans may have changed who was pregnant if they changed how pregnancies ended (Section I.B). Because the claims record women's health before they conceive, I test this directly by estimating the stacked design with each pregnancy's pre-pregnancy characteristics as the outcome.

A ban cannot change any one woman's health before conception, so a nonzero estimate means that the set of women who became pregnant changed.

The pregnant population's health before pregnancy changed little (Supplemental Appendix Table A13). After the bans, the comorbidity score changes by 0.4 percent ($p = 0.81$), the number of comorbid conditions by 0.7 percent, and the share with any condition by 0.4 percent ($p = 0.70$), and the last rules out increases larger than 2.6 percent. The Medicaid share of pregnancies, the most direct test of whether the added pregnancies came from women with low incomes, changes by 2.2 percent ($p = 0.28$), with an interval from -1.9 to 6.4 percent. The share of pregnancies to women under 25 rises by 11.2 percent ($p = 0.004$), a share that was already rising in ban states relative to protected states before the bans. Holding these characteristics fixed also changes the health estimates little, as they move by no more than a few tenths of a percent between the Raw and Controls columns of Table 2, so a change in the observed characteristics of who became pregnant would not explain them.

VII. Discussion

The *Dobbs* decision ended a constitutional right for women that had stood for half a century, and many expected the bans that followed to harm maternal health broadly, by delaying the management of complications and ending access to terminations for medical reasons. However, two features of the pre-*Dobbs* environment, both documented in outside work, are consistent with effects this much smaller than predicted. First, abortion care in states that subsequently banned abortion was already highly restricted, and those earlier restrictions had already caused harm, as restrictions on abortion providers raised hypertensive disorders of pregnancy (Gardner, 2024). The marginal effect of the bans was therefore potentially small. Second, even as restrictions came rapidly and repeatedly in the years leading up to *Dobbs*, the national environment loosened in some ways, as demand for abortion by telehealth rose during the COVID-19 pandemic (Aiken et al., 2020). In line with this work, insurance-billed abortion was already rare in ban states before the bans and continued to be rare after them (Section V.A). I find that the bans changed maternal health less than the warnings assumed, likely because much of what a ban restricts had already been restricted or could be obtained elsewhere.

How do these estimates compare with prior work? My interval for severe morbidity rules out increases larger than 5.6 percent, while the credible interval for pregnancy-associated mortality, a far rarer outcome, reaches 20.7 percent in Bell et al. (2026), so claims can exclude large harms that death records cannot. Prior work found large increases in severe complications within narrow

groups. The rise in sepsis among Texas women hospitalized with a second-trimester loss after SB8, described in Section I.B, is an increase of 55 percent. Among high-risk hospitalized pregnancies, a Texas time series with no comparison state associates the law with sustained increases in critical illness, sepsis, and infection, an association the same study does not detect across all hospitalized pregnancies (Chen et al., 2026). Because these groups are a small share of pregnancies, an effect of that size within them would move severe maternal morbidity among all insured pregnancies by far less than my bound. The Texas rise in sepsis was 1.6 percentage points among women hospitalized with a second-trimester loss. For a rise of that size to add the 15 cases per 10,000 pregnancies at the upper end of my interval, about 9 percent of pregnancies would need to end in a second-trimester loss, more than half of the 16 percent of ban-state pregnancies with an observed ending that ended in a miscarriage before the bans.

To make the magnitudes of my estimates concrete, consider 10,000 insured pregnancies in a ban state. Before the bans, about 264 of them would have involved severe maternal morbidity. The point estimate changes this by about -7.6, and the upper end of the 95 percent interval allows about 15 more. Without Texas, whose six-week ban came first, the interval reaches 11.8 percent, or about 34 more per 10,000 on the higher pre-ban rate of about 290 per 10,000 outside Texas (Supplemental Appendix Table A9). About 2,080 of the same pregnancies would have had a hypertensive disorder, and the interval allows at most about 71 more. For both outcomes, even the upper end is small next to how common they were before the bans.

However, these averages do not imply that the bans were harmless for every woman they reached and that any additional harm from the bans is an acceptable outcome. A pooled estimate spreads any harm across every insured pregnancy in a ban state. If the same added harm fell instead on the 2.5 percent of pregnancy endings that are ectopic, or on the smaller share that end in a second-trimester loss, the bound would allow the risk of severe morbidity in those groups to rise several times over. Estimated within losses, severe morbidity changes by 11.3 percent, with an interval that reaches 33.4 percent (Section V.D). Because losses are a small share of pregnancies and carry a lower rate of severe morbidity than deliveries, a rise of that size among them would move the pooled rate by only a few percent, inside its interval. The deaths and near-deaths documented in reporting and clinician interviews (Surana and Presser, 2025; Sabbath et al., 2024) are consistent with the small average estimates I report. My estimates speak to how common such harm was across a wide population, not to whether it occurred at all. They also measure physical health and care, not mental health, which studies find worsened among Texas mothers after its ban (Ali et al., 2026), or intimate partner violence, which rose after the bans (Dave et al., 2025).

Taken together with previous work, the results carry a clear lesson for policymakers and maternal health advocates. The debate over *Dobbs* has centered on the constitutional right to abortion, but for the women studied here, the restrictions that limited access came earlier. Restoring a federal right to abortion would lift the bans, but it would not by itself restore insurance coverage of abortion or the clinics those earlier restrictions closed. On its own, it may not meaningfully change the maternal health and care landscape for insured women in states that banned abortion.

Table 1—Characteristics of Pregnancies by State Policy and Period

	Ban States		Protected States	
	Before	**After**	**Before**	**After**
Pregnancies	140,836	121,531	437,625	295,385
Demographics, % Unless Noted				
Age, mean (SD)	29.01	28.84	30.35	30.18
	(6.40)	(6.82)	(6.23)	(6.57)
Under 25	26.7	31.0	19.3	22.6
Medicaid	42.4	37.1	53.4	56.3
Commercial	57.5	62.8	46.4	43.5
White	41.6	41.2	45.9	46.4
Black	19.2	16.2	14.0	14.4
Asian	1.9	1.8	3.9	3.7
Hispanic	8.7	14.0	23.8	26.5
First observed pregnancy	25.7	27.2	25.7	25.4
Health Before Pregnancy, % Unless Noted				
Hypertension, obesity, or substance use	23.6	25.5	24.0	27.5
Obstetric comorbidity score, mean (SD)	8.74	9.14	8.37	9.48
	(14.34)	(14.78)	(14.30)	(15.27)
Maternal Health, %				
Any adverse pregnancy outcome	28.6	29.2	26.1	25.5
Adjusted for characteristics	[29.5]	[30.2]	[25.7]	[25.2]
Severe maternal morbidity	2.8	2.6	2.7	2.8
Adjusted for characteristics	[2.9]	[2.6]	[2.7]	[2.7]
Hypertensive disorder	21.3	22.1	15.7	17.2
Adjusted for characteristics	[21.1]	[22.1]	[15.8]	[17.1]
Preterm birth (live births)	17.3	17.6	14.9	15.9
Adjusted for characteristics	[17.4]	[17.8]	[15.0]	[15.6]
Timing and Intensity of Care, % Unless Noted				
Length of pregnancy, days, mean (SD)	225.0	224.2	201.7	193.9
	(83.6)	(84.1)	(96.8)	(99.2)
Adjusted for characteristics	[223.7]	[223.6]	[201.3]	[195.3]
Days to first visit, mean (SD)	46.0	45.7	45.3	44.3
	(33.4)	(33.1)	(31.9)	(31.1)
Adjusted for characteristics	[47.7]	[48.3]	[44.3]	[43.8]
Prolonged delivery stay (live births)	23.3	24.2	24.3	25.9
Adjusted for characteristics	[23.2]	[23.6]	[24.6]	[25.8]
Number of visits, mean (SD)	16.2	16.3	16.3	16.3
	(10.9)	(10.8)	(12.5)	(13.1)
Adjusted for characteristics	[16.4]	[16.5]	[16.3]	[16.2]
Pregnancy Endings, % of Observed Endings Unless Noted				
Live birth	79.3	79.0	67.1	63.3
Adjusted for characteristics	[78.7]	[78.7]	[66.9]	[64.0]
Miscarriage	15.7	16.3	14.2	14.3
Adjusted for characteristics	[15.7]	[15.8]	[14.4]	[14.2]
Ending not observed (percent of all pregnancies)	6.4	5.7	4.9	5.8
Adjusted for characteristics	[5.6]	[5.0]	[5.2]	[6.0]
Insurance-billed abortion	1.2	0.9	15.2	18.6
Adjusted for characteristics	[1.8]	[1.6]	[15.2]	[18.1]

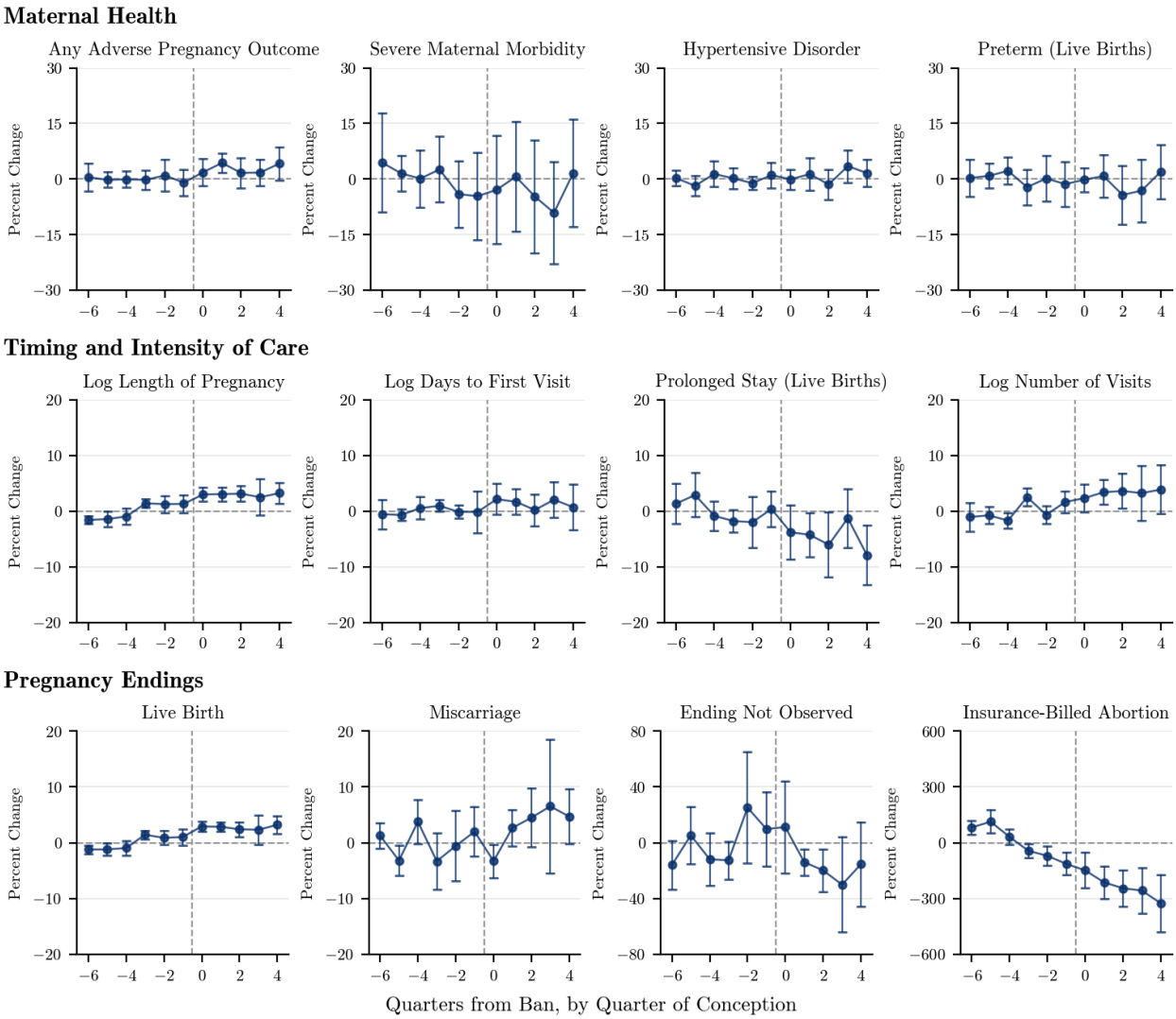
Notes: The table describes pregnancies started January 2020 through February 2024 in the continuous-enrollment sample. Ban states enacted a strict abortion ban; protected states never did. Before and after refer to pregnancies that began before or after the ban took effect in the woman's state, and after begins in June 2022 for protected states. Shares are percentages; age, the obstetric comorbidity score, the length of pregnancy, days to first visit, and number of visits are means with standard deviations (SD) beneath. Preterm birth and a prolonged delivery stay, a delivery admission longer than two days after a vaginal and four after a cesarean delivery, the 48 and 96 hours that federal law requires private plans to cover (Newborns' and Mothers' Health Protection Act, 42 U.S.C. 300gg-25), are percentages of live births; live birth, miscarriage, and insurance-billed abortion are percentages of observed endings; ending not observed is a percentage of all pregnancies. Hypertension, obesity, or substance use is a diagnosis of any of the three in the year before pregnancy. Each bracketed row adjusts the outcome for the individual controls of the main specification (the residual from a regression on them, recentered to the pregnancy-weighted average of the row above), so its gaps do not reflect those characteristics. Supplemental Appendix Table A6 reports normalized differences.

Figure 1. Raw Means of the Pregnancy Outcomes by Quarter of Conception



Notes: The figure plots means by calendar quarter of conception in the continuous-enrollment sample for the eleven states that banned at *Dobbs* (circles; first treated in the third quarter of 2022), Texas (squares; fourth quarter of 2021), and protected states (dashed line). A dotted vertical line for each cohort falls between its last untreated and first treated quarter, the earlier one for Texas. West Virginia, whose quarterly counts are small, and Indiana and South Carolina, whose post-ban window runs past the data, are omitted (Supplemental Appendix Table A5). Live birth, miscarriage, and insurance-billed abortion are percentages of pregnancies with an observed ending, preterm birth and a prolonged delivery stay of live births, and the other shares of all pregnancies; the length of pregnancy, the days to the first visit, and the number of visits are in logs. Quarters after the fourth quarter of 2023 are omitted because their pregnancies cannot all be followed to their ending.

Figure 2. Event Studies for the Pregnancy Outcomes



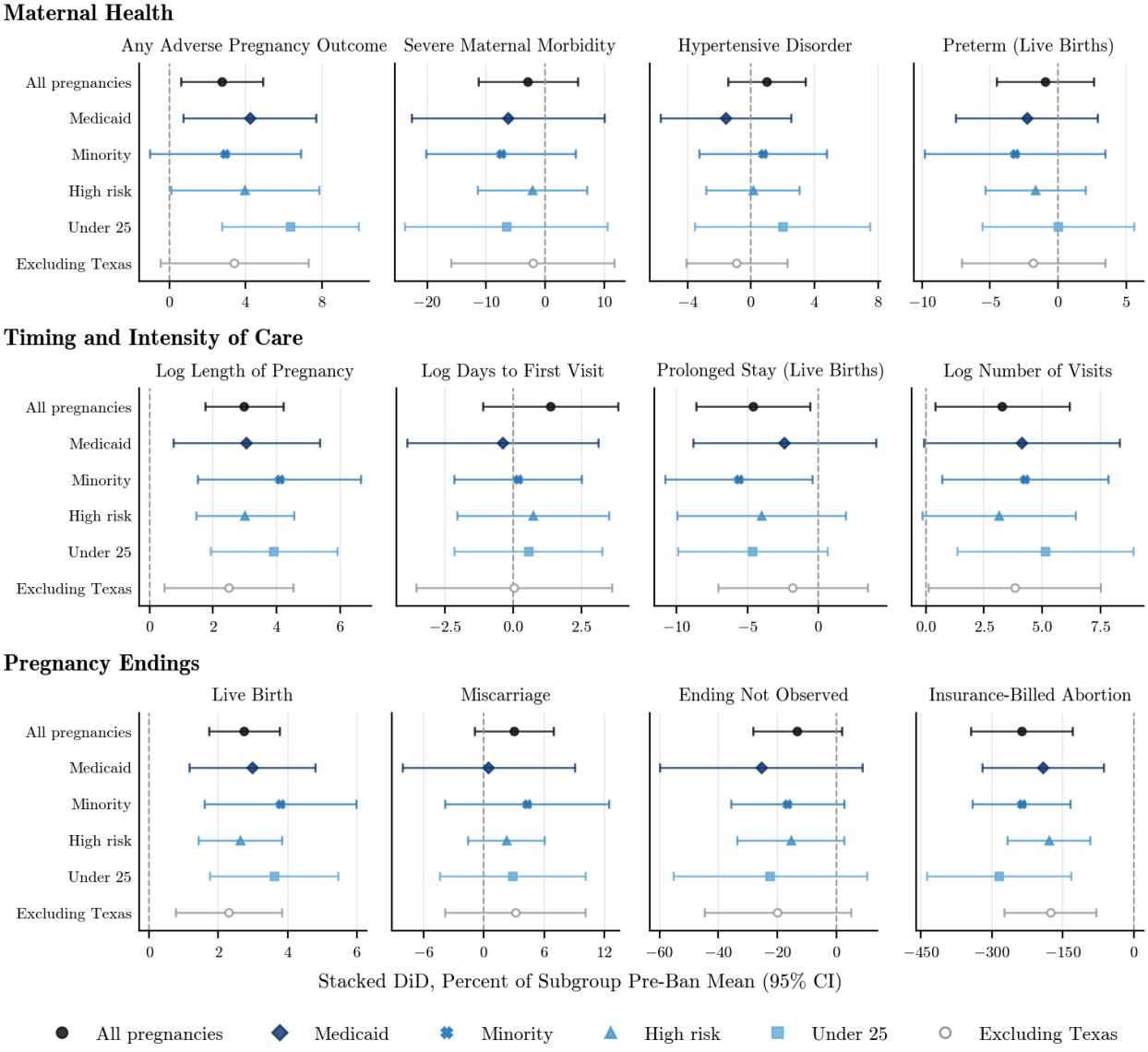
Notes: The figure plots stacked difference-in-differences event-study coefficients by quarter of conception relative to the ban, quarters −6 to 4, in the Controls specification and the continuous-enrollment sample. Bars show 95 percent confidence intervals from standard errors clustered by state. The vertical axis, percent change, is each coefficient as a percent of the outcome's ban-state pre-ban mean for shares, the scale of column 3 of Table 2, and 100 times the coefficient for the outcomes in logs. Each ban cohort is compared only with protected states, which never banned abortion, using the weights of Wing, Freedman and Hollingsworth (2024). The pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. Each vertical axis spans at least 20 percent on either side of zero, widened to a round number where an interval runs past it, and panels in a row share an axis unless one needs more than twice the narrowest. Table 2 reports pre-ban means and joint pre-trend tests.

Table 2—The Effect of Abortion Bans on the Pregnancy Outcomes

Outcome	Pre-Ban Mean (1)	Controls				Raw	
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Pre-Trend p-Value (5)	Estimate (SE) (6)	Percent of Mean (7)
Maternal Health							
Any adverse pregnancy outcome	0.285	0.008 (0.003)	2.8	[0.6, 4.9]	0.98	0.009 (0.003)	3.2
Severe maternal morbidity	0.026	−0.001 (0.001)	−2.9	[−11.4, 5.6]	0.86	−0.001 (0.001)	−2.4
Hypertensive disorder	0.208	0.002 (0.003)	1.0	[−1.5, 3.4]	0.48	0.003 (0.002)	1.4
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]	0.29	−0.001 (0.003)	−0.8
Timing and Intensity of Care							
Log length of pregnancy	5.272	0.030 (0.006)	3.0	[1.8, 4.2]	<0.001	0.033 (0.008)	3.3
Log days to first visit	3.616	0.014 (0.012)	1.4	[−1.1, 3.8]	0.12	0.014 (0.013)	1.4
Prolonged delivery stay (live births)	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]	0.03	−0.007 (0.004)	−3.2
Log number of visits	2.496	0.033 (0.014)	3.3	[0.4, 6.2]	<0.001	0.038 (0.016)	3.8
Pregnancy Endings							
Live birth	0.789	0.022 (0.004)	2.7	[1.7, 3.8]	<0.001	0.024 (0.005)	3.0
Miscarriage	0.162	0.005 (0.003)	3.0	[−0.9, 6.9]	<0.001	0.003 (0.004)	1.7
Ending not observed	0.062	−0.008 (0.005)	−13.3	[−28.3, 1.6]	0.001	−0.009 (0.005)	−14.1
Insurance-billed abortion	0.011	−0.025 (0.006)	−237.1	[−344.5, −129.6]	<0.001	−0.024 (0.005)	−232.0

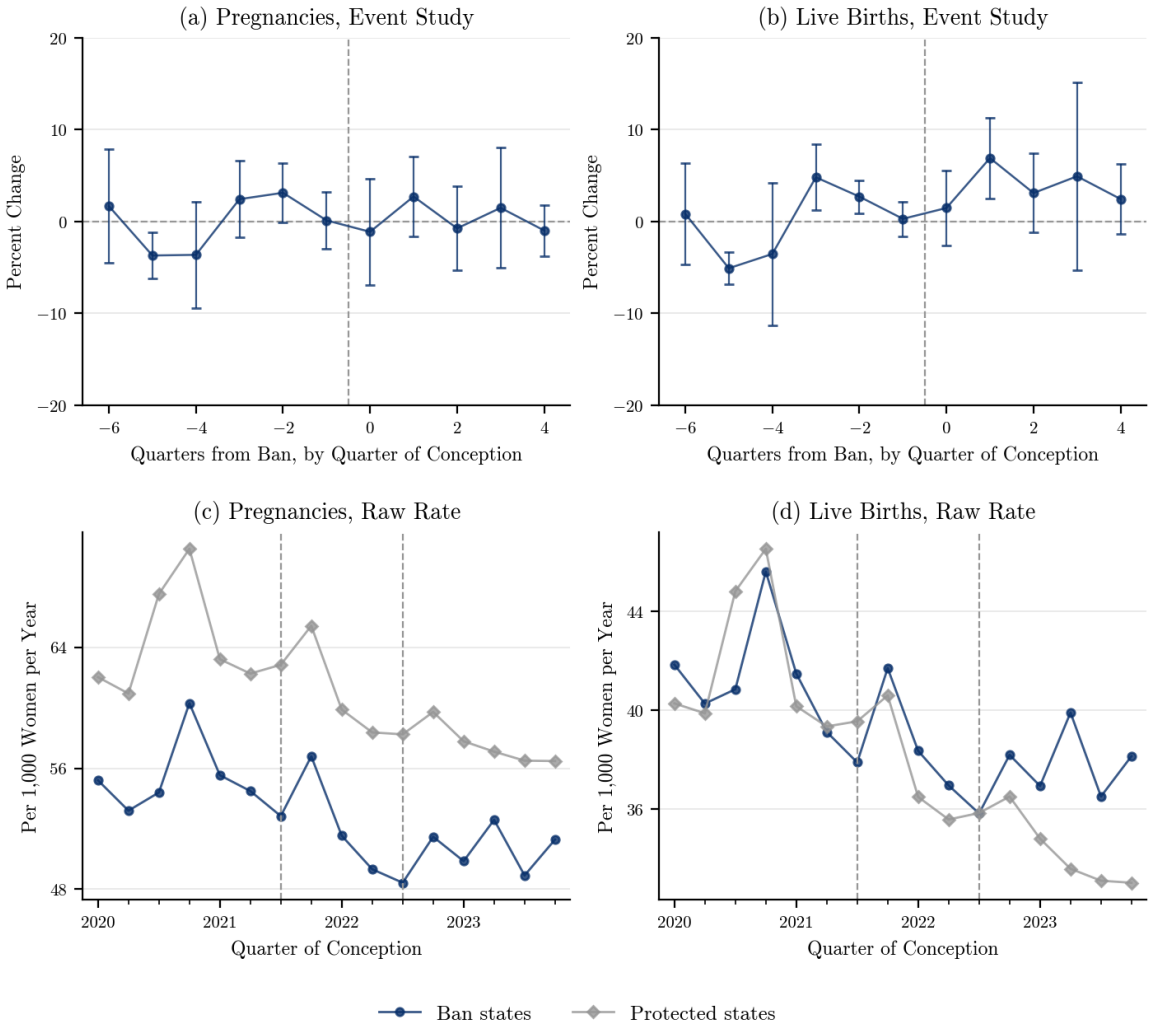
Notes: Estimates use the event-study sample (Section IV), the 818,245 continuous-enrollment pregnancies conceived in the eleven quarters around each cohort's ban (fewer for outcomes defined for observed endings or live births). Column 1 is the ban-state mean in event quarters −6 to −1. For severe maternal morbidity, column 2 is −7.6 (11.1) per 10,000 pregnancies. Columns 1, 2, and 6 are in each outcome's units (proportions or logs); preterm birth and a prolonged delivery stay (Table 1) are shares of live births, live birth, miscarriage, and insurance-billed abortion are shares of observed endings, and ending not observed is a share of all pregnancies. Under Controls, column 2 is the stacked difference-in-differences estimate of the post-ban quarters against the average of quarters −6 to −1, with event-time and sub-experiment by state fixed effects, the weights of Wing, Freedman and Hollingsworth (2024), and the individual controls (age and its square, six race and ethnicity groups, Medicaid, first observed pregnancy, multiple gestation, and the comorbidity score of Leonard et al. 2020), with its standard error (SE), clustered by state, in parentheses; column 3 is that estimate as a percent of the pre-ban mean (100 times the coefficient for outcomes in logs), column 4 its 95 percent confidence interval (CI) on the same scale, in brackets, and column 5 the p-value of the joint test that the pre-period event-study coefficients are zero. Under Raw, columns 6 and 7 repeat columns 2 and 3 without the individual controls. The insurance-billed abortion estimate exceeds 100 percent of its mean because insured abortion rose in protected states (Figure 1).

Figure 3. Heterogeneity: Percent of Each Subgroup's Pre-Ban Mean



Notes: Each marker is a separate stacked difference-in-differences (DiD) regression in the named group, as a percent of that group's pre-ban mean (100 times the coefficient for outcomes in logs), with a bar for its 95 percent confidence interval (CI) from standard errors clustered by state, in the Controls specification and the continuous-enrollment sample. The all-pregnancies row repeats Table 2, and each panel has its own scale. Medicaid pregnancies have a Medicaid payer in the month the pregnancy started; minority women have a Black race code or a Hispanic ethnicity code; high-risk women have at least one condition in the obstetric comorbidity index before pregnancy; and excluding Texas removes the Texas cohort. The dashed vertical line marks zero, and Supplemental Appendix Table A9 reports the estimates.

Figure 4. Fertility: Event Studies and Raw Rates



Notes: Panels (a) and (b) plot stacked difference-in-differences event-study coefficients for the pregnancy and live-birth rates on the panel of enrolled women aged 15 to 45 by state and quarter, with event-time and sub-experiment by state fixed effects and weights for enrolled women; bars show 95 percent confidence intervals from standard errors clustered by state. The pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. Their vertical axis, percent change, is each coefficient as a percent of the ban-state pre-ban rate, as in Supplemental Appendix Table A12, and is drawn as in Figure 2. Panels (c) and (d) plot the raw rates by quarter of conception in ban states (circles) and protected states (diamonds) in the continuous-enrollment sample, monthly rates averaged to quarters; dashed lines mark the quarter in which Texas Senate Bill 8 (the third quarter of 2021) and the *Dobbs* bans (the third quarter of 2022) took effect. All rates are per 1,000 enrolled women per year, and Supplemental Appendix Figure A1 shows the episode-enrolled sample.

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Supplemental Appendix

It Was Already Dobbs: Abortion Bans and Maternal Health

Appendix

Section A1 documents the timing of the bans, how each state is classified, and the legal and policy sources. Section A2 describes how I build and date pregnancy episodes. Section A3 describes the construction of the samples and their pre-period balance. Section A4 compares the continuous-enrollment sample with the episode-enrolled sample. Section A5 reports the subgroup estimates, and Section A6 breaks the adverse-outcome composite into its components. Section A7 reports the fertility estimates and the composition of the pregnant population. Section A8 documents the stacked design and compares it with two-way fixed effects, other reference periods, and synthetic difference-in-differences, and Section A9 the close-to-care design. Section A10 defines the outcomes.

A1. Ban Timing, State Classification, and Policy Sources

Policy sources

Table A1 lists the statute behind each ban. The *Dobbs* decision (2022) overturned *Roe v. Wade* (1973). In Texas, SB8 (Texas Legislature, 2021*b*) took effect on September 1, 2021, and because private lawsuits rather than state officials enforced it, there was no official a federal court could enjoin to stop it (*Whole Woman's Health v. Jackson*, 2021). The state's trigger law (Texas Legislature, 2021*a*) took effect on August 25, 2022 (Texas State Law Library, 2026). After *Dobbs*, federal courts allowed the previously enjoined bans of Alabama (*Robinson v. Marshall*, 2022) and Georgia (*SisterSong v. Governor of Georgia*, 2022) to take effect. The author of the Texas law and Kentucky Senate Republicans have each described the reach of their state's health exceptions (Hughes, 2024; Kentucky Senate GOP, 2025), and reporting on Kate Cox's case in Texas documents a woman denied a termination despite a serious health risk (Klibanoff, 2023). Further investigative reporting documents pregnant women in Texas who died after emergency care was delayed (Presser and Surana, 2024; Jaramillo and Surana, 2024). The Texas sepsis figures in Section I.B come from hospital discharge records analyzed by Presser et al. (2025), whose method is described in Suozzo, Chou and Presser (2025). Among protected states, Illinois widened protections and coverage after

Dobbs (Office of Governor JB Pritzker, 2023), and California widened protections for providers and patients (Office of Governor Gavin Newsom, 2023).

A2. Constructing and Dating Pregnancy Episodes

Endings

To find the ending of a pregnancy, I flag every claim carrying a diagnosis, procedure, or diagnosis-related-group code for a live birth, stillbirth, molar pregnancy, ectopic pregnancy, induced abortion, or spontaneous abortion, each as a separate ending type. Codes that indicate the ending of a pregnancy may span several days, for example several days of labor and the postpartum care that follows, so I link ending codes that fall within 30 days of the previous one into a single ending, dated at the earliest claim. Where an ending carries codes for more than one type, I apply the hierarchy that Moll et al. (2021) validated against physician-adjudicated records, which orders endings by how reliably each is coded, from live birth, stillbirth, molar pregnancy, ectopic pregnancy, and induced abortion to spontaneous abortion (miscarriage). Because the abortion codes include the billing codes for mifepristone and misoprostol, which also treat miscarriage, and the hierarchy ranks induced abortion above miscarriage, a miscarriage managed with these medications can be counted as an abortion, so the classification errs toward counting too many abortions rather than too few. Additionally, because women may use the health care and insurance systems intermittently throughout pregnancy, especially for an early miscarriage or an abortion paid outside insurance, some endings imply a pregnancy that began implausibly soon after the woman's previous one ended. To account for this, I treat an ending whose estimated start falls within the recovery period after her previous ending, 24 weeks after a birth or stillbirth and 60 days after any other ending, as a continuation of the earlier pregnancy rather than a new one, and I exclude it as a separate ending.

Pregnancy start

I date each pregnancy from the first day of the estimated last menstrual period, the clinical convention from which gestational age is counted (American College of Obstetricians and Gynecologists, 2017), and call this date the pregnancy start. I largely rely on the Z3A diagnosis code, which records the gestational week at a visit (American Academy of Professional Coders, 2024).[1] For each Z3A code in the 300 days before the ending, I subtract the recorded weeks from the date of service, and I take the median of these estimates, assigning four weeks to codes below eight weeks,

[1]Where an embryo transfer is observed, I set the start 14 days before it, as Matcho et al. (2018) date starts from fertility procedures. Because the transfer of a day-3 or day-5 embryo corresponds to 17 or 19 days of gestation (American College of Obstetricians and Gynecologists, 2017), this places these starts up to five days late.

as the validated algorithm does (Chomistek et al., 2023). In my analysis sample, Z3A codes date 87 percent of ban-state and 75 percent of protected-state pregnancies (Table A3). Against last-period dates that an epidemiologist adjudicated from medical records, this method has a median absolute error of four days, with 65 percent of estimates within a week (Chomistek et al., 2023). However, its accuracy varies by ending. For example, 89.7 percent of premature deliveries fall within seven days, but only 34.3 percent of spontaneous abortions do, because early losses often end before the first visit at which a Z3A code is recorded. These facts give the first of two reasons I measure event time in quarters rather than months (Section IV), as miscarriages alone end about 16 percent of ban-state pregnancies with an observed ending before the bans, so monthly event time would add noise from dating error. Second, pooling to quarters estimates each event-time coefficient on more pregnancies, at the cost of making changes within a quarter harder to detect.

I date episodes without a Z3A code from the earliest test performed at a known gestational age, adapting the priority orders in Moll et al. (2021) and Matcho et al. (2018).[2] Failing all of these, I assign the ending-type defaults of Chomistek et al. (2023), which its authors drew from the literature and obstetrician input (39 weeks for a singleton live birth, 36 for a multiple birth, 28 for a stillbirth, 10 for an abortion of either type, and 8 for ectopic and molar pregnancies). Table A3 reports the share of episodes dated by each source.

Pregnancies without an observed ending

Some pregnancies carry pregnancy care but no ending code. In my core analysis sample, they are 6.2 percent of ban-state pregnancies conceived in the six quarters before each state's ban (Section II.D). To account for these pregnancies, I build episodes with no observed ending from pregnancy-indicating claims (for example, prenatal visits, gestational-age codes, and diagnoses of adverse pregnancy outcomes) that fall outside every other observed pregnancy for a woman,[3] as Sarayani et al. (2020) do for pregnancies with unknown outcomes. I group these claims into clusters, starting a new cluster whenever more than 60 days separate two claims, similar to the two-month separation between pregnancies that Ailes et al. (2016) require, and a cluster with claims on at least two dates becomes an episode. Because nothing in the claims marks how such a pregnancy ended, I end the episode on the date of its last pregnancy-indicating claim. I date its start from its

[2] My order is nuchal translucency ultrasound, cell-free DNA screening, and maternal serum screening, then the detailed fetal anatomy ultrasound and glucose tolerance testing. Moll et al. (2021) first rank gestational-age codes by the service that carries them, then date pregnancies without such a code from nuchal translucency ultrasound, chorionic villus sampling, cell-free DNA screening, and later the glucose tolerance test. I instead date from the median of all Z3A codes, as Chomistek et al. (2023) validate, and add maternal serum screening, which Matcho et al. (2018) use. I exclude hCG tests from the serum-screening tier, because they are drawn early in pregnancy to work up a suspected ectopic pregnancy or loss (American College of Obstetricians and Gynecologists, 2018*a*,*b*) rather than at the gestational age that tier assumes.

[3] An observed pregnancy runs from 90 days before its start through its 42-day postpartum window.

latest Z3A code or, if it has none, eight weeks before its first visit, close to the median time from the last menstrual period to the first pregnancy-care claim among deliveries in insurance claims (Sarayani et al., 2020). Finally, a cluster whose last claim falls within 200 days before an observed ending is likely early care for that same pregnancy rather than a separate one, so I exclude it.[4]

A3. Sample Construction and Descriptives

Additional restrictions

The pregnancy episode construction described in Section II.C and Section A2 yields 10,635,041 pregnancies for women aged 15 to 45, 1,848,589 of them to continuously enrolled women. Among those, I keep pregnancies in the 15 ban states (the 13 treated states, Indiana, and South Carolina) and 25 protected jurisdictions (1,576,819) and pregnancies that started in 2020 or later (1,135,994), because some pregnancies observed in my window started in 2019 and continued into 2020. I require the pregnancy to start no later than February 2024 so that a typical 40-week pregnancy can be followed to its ending (996,163). Because an implied gestation shorter than 28 or longer than 315 days likely reflects a coding or dating error, I exclude these observed endings, about 0.1 percent of the remaining pregnancies. The continuous-enrollment sample contains 995,377 pregnancies. Table A4 reports counts after each step.

The pre-pregnancy year

I exclude pregnancies that start in 2019 (Section II.D). For pregnancies that start in 2020, part of the pre-pregnancy year falls in 2019, before the enrollment requirement begins, so their baseline health measures draw on whatever 2019 claims the woman has. Of these 2020 pregnancies, 81 percent were also enrolled for the whole year before the pregnancy, from 72 percent of those that started in the first quarter to 91 percent in the fourth. The share is similar in ban states (83 percent) and protected states (81 percent), so a shorter look-back does not fall more on one side of the comparison of baseline health.

Provider locations

Claims locate care only through the registered address of the provider's NPI, so in my core analysis I assign treatment by the woman's state of residence in the enrollment file. Of all claim lines, 25.5 percent name no provider, and 24.3 percent are billed under an organizational NPI,

[4]This 200-day window is comparable to the 168 to 182 days that Matcho et al. (2018) require between an earlier outcome and a live birth.

whose address may be a billing office or a system headquarters, not the place of care.

Pre-pregnancy comorbidity score

The conditions of the obstetric comorbidity index (Section II.D) include pulmonary hypertension, cardiac and renal disease, chronic hypertension, pre-existing diabetes, autoimmune and bleeding disorders, anemia, asthma, and mental health and substance use disorders, each measured in the year before conception. I omit placenta accreta, because it is diagnosed during pregnancy. The score weights each pre-pregnancy condition by the points Leonard et al. (2020) estimated for severe maternal morbidity (Table A21). I also report the number of conditions and an indicator for any condition, measures that do not depend on the weights (Section VI.B).

A4. Comparison with the Episode-Enrolled Sample

The episode-enrolled sample requires enrollment from twelve months before the pregnancy start through three months after the ending, so it keeps women who move between insurers or plans outside their pregnancies; it contains 2,416,491 pregnancies. The main analysis does not use it, because its composition changes over the period for reasons the bans likely did not cause. First, insurers enter and leave the data, and because plans can be regional, each entry or exit can shift who is in the sample. Second, its requirement of coverage through three months after each pregnancy ends means that its membership after mid-2022 is set by the end of the pandemic-era Medicaid continuous-enrollment provision on March 31, 2023, after which states began removing women from Medicaid (Tolbert and Ammula, 2023). In this sample, the resulting loss of members is concentrated in ban states. Between conceptions in the second quarter of 2022 and the third quarter of 2023, the number of episode-enrolled pregnancies changes by -42 percent in ban states and -22 percent in protected states, and the Medicaid share of ban-state pregnancies in that sample falls from 61.5 to 45.9 percent, while the protected-state share holds near 60.4 percent. Over the same quarters, the number of continuous-enrollment pregnancies changes by 2 percent in ban states and -2 percent in protected states. Figure A1 shows the birth and pregnancy rates in the episode-enrolled sample, where the ban-state birth rate falls to the protected-state rate after 2022, as the end of Medicaid continuous enrollment would predict in a sample whose membership it sets.

Table A8 reports the difference-in-differences estimates in both samples under the Controls specification, and Figure A2 overlays their event studies. The maternal health estimates are similar in the two samples. In the episode-enrolled sample, the adverse-outcome composite changes by 1.3 percent and severe maternal morbidity by 0.5 percent, ruling out increases larger than 4.9 and 6.4

percent. However, the pregnancy endings differ. In this sample the shift runs toward miscarriages rather than live births, as the miscarriage share rises 10.1 percent ($p = 0.03$) and the live-birth share changes by 1.0 percent ($p = 0.59$). Because the end of Medicaid continuous enrollment changes who remains in this sample, I cannot tell how much of that difference reflects the bans rather than the sample's changing composition.

A5. Subgroup Estimates

Table A9 reports the stacked estimates behind Figure 3 for each subgroup of Section V.D. Table A10 repeats the maternal health and care outcomes within live births and within losses. Because the bans changed how pregnancies ended, these within-ending estimates describe the pregnancies in each group rather than isolate effects of the bans.

A6. Components of the Adverse-Outcome Composite

The adverse-outcome composite equals one if a pregnancy has any of five diagnoses (Section II.E). Table A11 estimates each on its own and Figure A3 shows their event studies. Hypertensive disorders, reported in Table 2, are not part of the composite.

A7. Fertility and the Composition of the Pregnant Population

Table A12 reports the birth and pregnancy rates of Section VI.A for all enrolled women and for each subgroup, and Table A13 reports the stacked estimates for the pre-pregnancy health and demographic measures discussed in Section VI.B.

A8. Alternative Estimators

The stacked design and two-way fixed effects

The event-study sample is the continuous-enrollment pregnancies conceived in the eleven quarters around each cohort's ban, from the second quarter of 2020 through the fourth quarter of 2022 for Texas, from the first quarter of 2021 through the third quarter of 2023 for the *Dobbs* cohort, and from the second quarter of 2021 through the fourth quarter of 2023 for West Virginia. Table A4 reports the number of distinct pregnancies in this event-study sample, 818,245, and Table A14 the pregnancies in each sub-experiment. Because a control pregnancy enters every sub-experiment whose window covers it, the stacked regression has more observations than distinct pregnancies. The two-way fixed effects coefficients in Figure A4 lie close to the stacked ones for every outcome and quarter, so the stacked weights do not drive the estimates.

Reference quarter

The pooled estimates in Table 2 compare the post-ban quarters with the average of quarters −6 to −1 (Section IV). Table A15 repeats them against quarter −2 alone. Figure A5 draws the event studies against the same quarter. I use quarter −2 rather than quarter −1, the reference in Wing, Freedman and Hollingsworth (2024), as a compromise, because quarter −1 is the most exposed to the ban, since pregnancies conceived then are still under way when it takes effect and Texas and West Virginia banned inside it. The post-ban quarters are the same, so each estimate moves only by the difference between the reference periods. Against quarter −2, the suggestion of harm in the composite weakens. It falls from 2.8 to 1.8 percent of the pre-ban mean ($p = 0.40$), with an interval that rules out increases larger than 5.9 percent, and its pre-trend test, which does not depend on the reference, is in Table 2. Hypertensive disorders move the other way, to 2.1 percent ($p = 0.06$), and preterm birth changes by −1.1 percent. Severe maternal morbidity changes by 1.2 percent, with an interval that rules out increases larger than 12.1 percent, and the fall in prolonged delivery stays shrinks to 2.6 percent ($p = 0.21$). The number of visits rises 4.0 percent ($p = 0.001$).

Synthetic difference-in-differences

If protected states are a poor comparison for ban states, the stacked estimates could reflect differences in trends rather than the bans. Synthetic difference-in-differences (Arkhangelsky et al., 2021) addresses this by choosing weights on protected states and on pre-ban quarters so that the weighted protected-state path matches the ban states' path before the bans. I estimate it on state by conception-quarter means, with 38 states, 13 of them treated, and 16 quarters, and placebo standard errors (Table A16). Insurance-billed abortion falls 213.9 percent of the pre-ban mean ($p = 0.01$), and the maternal health estimates are close to the stacked ones, with wider intervals. The adverse-outcome composite changes by 2.0 percent (95 percent confidence interval −7.7 to 11.7 percent) and severe maternal morbidity by −4.3 percent (−32.6 to 24.0 percent). Because synthetic difference-in-differences uses state means rather than pregnancies, its intervals are wider than the stacked ones, and it does not distinguish the rises in live births and in the composite from zero.

Table A17 repeats the fertility estimates of Table A12 with synthetic difference-in-differences on state by quarter rates, and Figure A6 compares the event studies. Synthetic difference-in-differences puts the change in the birth rate at 6.2 percent (95 percent confidence interval −2.8 to 15.2 percent) and in the pregnancy rate at 1.9 percent (−3.7 to 7.5 percent).

A9. Close-to-Care ZIP Code Design

As a check that does not rely on protected states, I implement a close-to-care ZIP-code design that restricts the sample to pregnancies in ban-state ZIP codes and compares women by their distance to emergency care in protected states. Because every woman in this sample faced a ban, the design measures how much closer protected-state care softened the ban, not the ban's full effect. A pregnancy is close to care, the comparison group, when the centroid of its assigned ZIP code lies within 50 miles of the ZIP-code centroid of the nearest general acute care hospital or emergency medicine provider (an organizational, Type 2, NPI in NPPES) in a protected state. Restrictions on abortion care likely bind less for women who live closer to care outside their state, because crossing the border for care costs them less, especially in an emergency. Pregnancies farther than 50 miles from protected-state care form the treated group.

I first assign each pregnancy a ZIP code, because the data do not record a woman's five-digit ZIP code. For each pregnancy I take the modal practice ZIP code, from NPPES, of the individual (Type 1) providers the woman saw in the 365 days before her pregnancy started, and use it as her assigned ZIP code. Assigning the ZIP code pregnancy by pregnancy rather than once per woman avoids a stale location for women who move between pregnancies, and using individual providers avoids placing a woman at a hospital system's headquarters. Dench et al. (2025) find that total bans raised net out-migration from ban states, and because I assign each pregnancy's location from the providers the woman saw before it and require continuous enrollment, a woman who moves is placed where she received care before that pregnancy rather than where she lived at the start of the window. Treatment timing and the state fixed effects follow the state of the assigned ZIP code, and I cluster standard errors by assigned ZIP code, the unit at which treatment is assigned. The design only works if the assigned ZIP code is a reasonable proxy for residence. Table A18 reports placement diagnostics by sub-sample. In every sub-sample, a majority of the visits with a provider ZIP code fall in the modal ZIP code, and most fall within its three-digit ZIP area, so the assigned ZIP code likely describes where the woman gets care. In the full close-to-care sample, before the continuous-enrollment restriction, 12.1 percent of pregnancies lie within 50 miles of protected-state care, and the mean distance from the assigned ZIP code to that care is 248 miles. The assigned ZIP code is not an exact residence but a stable location that supports a contrast between women close to and far from care.

Figure A7 maps the close-to-care classification across all lower-48 ZIP codes. Ban-state ZIP codes within 50 miles of protected-state care identify the close-to-care (comparison) group, ban-state ZIP codes farther than 50 miles form the treated group, and protected-state ZIP codes near the ban

states form the natural destinations. ZIP codes in states whose abortion policy was ambiguous during the window are white and excluded from both groups. The continuous-enrollment close-to-care sample contains 235,305 pregnancies, and the full close-to-care sample spans 5,462 assigned ban-state ZIP codes. Sample descriptives for the close-to-care groups appear in Table A19.

Table A20 reports the close-to-care estimates and Figure A8 their event studies. The design is less precise than the main one, and its maternal health estimates are close to or below the main ones. The adverse-outcome composite changes by −3.0 percent far from protected-state care relative to near it, preterm birth by −9.0 percent, and severe maternal morbidity by −26.6 percent, none distinguishable from zero. Insurance-billed abortion, already rare on both sides of the 50-mile line, changes by −56.9 percent, with an interval from −141.4 to 27.5 percent. Only the share of pregnancies with no observed ending moves clearly, falling 32.4 percent ($p = 0.02$). These estimates are consistent with the main results in a design that does not rely on protected states, although their intervals are too wide to rule out effects that the main design rules out. The design does not show the shift toward live births (−0.4 percent).

A10. Outcome Definitions

Severe maternal morbidity uses the 16 diagnosis indicators in the definition of the Centers for Disease Control and Prevention (Section II.E). The definition adds five procedure indicators: transfusion, hysterectomy, ventilation, temporary tracheostomy, and conversion of cardiac rhythm. I do not use them, because procedure codes are recorded mainly on facility claims and their completeness varies by payer and setting, and because transfusion is the most common source of false positives in chart validation (Main et al., 2016). I measure preterm birth from gestational age at delivery for pregnancies dated by Z3A codes or embryo transfer (Leonard et al., 2020, Appendix 2), and otherwise from the preterm-labor diagnosis. Insurance-billed abortion counts an ending with any abortion-related code. The codes are ICD-10-CM Z33.2 (encounter for elective termination), O04, and O07; CPT 59840, 59841, 59850 to 59852, and 59855 to 59857; ICD-10-PCS 10A0; and HCPCS S0190, S0191, S0199, S2260, and S2265 to S2267.

Table A1—State Abortion Policy Classification

State	Ban in Force	Law	First Treated Quarter
Ban States (N = 15)			
Texas	September 1, 2021	Six-week ban (SB8)	Fourth quarter of 2021
Alabama	June 24, 2022	Total ban (HB 314, 2019)	Third quarter of 2022
Arkansas	June 24, 2022	Trigger ban	Third quarter of 2022
Georgia	July 20, 2022	Six-week ban (HB 481)	Third quarter of 2022
Idaho	August 19, 2022	Six-week ban; trigger ban August 25	Third quarter of 2022
Kentucky	June 24, 2022	Trigger ban	Third quarter of 2022
Louisiana	June 24, 2022	Trigger ban	Third quarter of 2022
Mississippi	July 7, 2022	Trigger ban	Third quarter of 2022
Missouri	June 24, 2022	Trigger ban	Third quarter of 2022
Oklahoma	May 3, 2022	Civil bans; trigger ban June 24	Third quarter of 2022
South Dakota	June 24, 2022	Trigger ban	Third quarter of 2022
Tennessee	June 28, 2022	Six-week ban; total ban August 25	Third quarter of 2022
West Virginia	September 16, 2022	Total ban (HB 302)	Fourth quarter of 2022
Indiana	August 21, 2023	Total ban (SEA 1)	Fourth quarter of 2023
South Carolina	August 23, 2023	Six-week ban (S. 474)	Fourth quarter of 2023

Protected States and DC (N = 25)
Alaska, California, Colorado, Connecticut, Delaware, District of Columbia, Hawaii, Illinois, Kansas, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Rhode Island, Vermont, Virginia, Washington

Excluded States (N = 11)
Arizona, Florida, Iowa, Nebraska, North Carolina, North Dakota, Ohio, Pennsylvania, Utah, Wisconsin, Wyoming

Notes: The date is the first date a ban at six weeks of gestation or earlier was in force. The analysis gives every state in a cohort the cohort's first fully treated quarter (last column). Table A2 gives the reasons for each state whose legal date differs from its cohort's and for each excluded state. Three bans were also briefly in force in 2022: South Carolina's six-week ban from late June to August 17, West Virginia's 1849 ban from June 24 to July 20, and Indiana's ban from September 15 to 22. The West Virginia episode falls in that state's pre-ban quarters, which the main estimates dilute by comparing with the average of all six, and Indiana and South Carolina are excluded from the pregnancy estimates. Protected states kept abortion legal through at least 22 weeks of gestation throughout the window. Excluded states had bans that were enjoined, partial, or changing during the window. Classification follows Dench, Pineda-Torres and Myers (2024). Sources: Dench, Myers and Pineda-Torres (2025, Table A.1), Nash and Guarnieri (2022), Curhan (2026), state statutes, and court orders.

Table A2—Classification Decisions for States with Contested Timing or Status

State	Group and Timing in the Analysis	Reason
Ban States		
Texas	Ban; first treated in the fourth quarter of 2021	SB8 took effect September 1, 2021, and was never enjoined, because its private enforcement left no official to sue; clinics stopped providing abortion after about six weeks that day. The August 2022 trigger ban extended a ban already in force on the ground.
Oklahoma	Ban, *Dobbs* cohort; third quarter of 2022	Civilly enforced bans took effect in May 2022, weeks before *Dobbs*, but the ban regime was incomplete until then: enforcement rested on private suits, and the criminal trigger ban took effect only with *Dobbs*. Coding Oklahoma with the *Dobbs* cohort places May and June 2022 in its reference quarter, so any early effect raises the reference level and pulls its estimate toward zero rather than away from it.
Georgia, Idaho, Mississippi	Ban, *Dobbs* cohort; third quarter of 2022	Each ban took effect between July 7 and August 25, 2022, inside the cohort's first post quarter. That quarter is only partly treated for these states, which attenuates the estimate at event time 0 (Section IV); every later quarter is fully treated, apart from November 15 to 23, 2022, when a state court blocked Georgia's ban.
Kentucky, Louisiana	Ban, *Dobbs* cohort; third quarter of 2022	State courts blocked the trigger bans for parts of July 2022, but each ban was in force from late June, again from August 2022, and for the rest of the window. The interruptions fall inside the partly treated first post quarter.
West Virginia	Ban, own cohort; fourth quarter of 2022	Providers stopped under the 1849 statute from June 24 until a court enjoined it on July 18, 2022, and resumed until the total ban (HB 302) took effect September 16, 2022. The analysis dates the ban from the statute that remained in force; the three-week pause falls in West Virginia's reference quarter and, if it mattered, biases its estimate toward zero.
Indiana	Ban from the fourth quarter of 2023; not in the stacked pregnancy estimates	SEA 1 was in force for one week in September 2022 and then enjoined, and abortion remained available until August 21, 2023. Indiana has no complete post-ban window in the pregnancy data, so it does not enter the stacked pregnancy estimates.
South Carolina	Ban from the fourth quarter of 2023; not in the stacked pregnancy estimates	A six-week ban was in force from June 27 to August 17, 2022, before the state supreme court blocked it; abortion was then legal to about 22 weeks until August 23, 2023. South Carolina has no complete post-ban window in the pregnancy data, so it does not enter the stacked pregnancy estimates.
Protected States		
Kansas	Protected	Abortion is legal to 22 weeks, before viability, but the state constitution protects it and voters rejected an amendment to remove that protection in August 2022. The 22-week limit binds fewer than 1 percent of abortions (Kortsmit et al., 2023), and more than three-quarters of abortions in Kansas in 2023 were to nonresidents, most from neighboring ban states (Conlon, 2024).
Michigan	Protected	The 1931 ban was enjoined on May 17, 2022, before *Dobbs*, and was never enforced afterward; Proposal 3 added the right to the state constitution in November 2022.
Excluded States		
Wisconsin	Excluded	Providers stopped under the 1849 statute from June 2022, a ban on the ground, but they resumed in September 2023 after a court ruling. Treating Wisconsin as a ban state would place a reversal inside its post-period, and treating it as protected would place fifteen months of no access inside a control.
Ohio, Utah, Arizona	Excluded	Each had a near-total or six-week ban in force for days to months in 2022 and a gestational limit of 15 to 18 weeks, or a later reversal, for the rest of the window, so neither group describes it.
Florida, Nebraska, North Carolina, Iowa	Excluded	Each moved from legal abortion to a 15-week, 12-week, or six-week limit during the window, a partial restriction that changed within the period rather than a ban or protected access throughout.
North Dakota	Excluded	The trigger ban was blocked by a state court, but the state's only clinic moved to Minnesota in August 2022, so legal status and access on the ground diverged; a near-total ban (SB 2150) then took effect on April 24, 2023.
Wyoming	Excluded	The trigger ban was in effect for a day in July 2022 and then enjoined, with further bans litigated through November 2024.
Pennsylvania	Excluded	Abortion was legal to 24 weeks throughout, but the state kept one of the most restrictive regimes among states where it remained legal: a 24-hour waiting period with state-directed counseling, parental consent for minors, no public coverage beyond the federal Hyde exceptions, and a requirement that clinics meet ambulatory surgical facility standards, a targeted regulation of abortion providers. Placing it among the protected states would put a restrictive regime in the comparison group, which would bias the estimates toward zero.

Notes: The classification in Table A1 is fixed in the analysis code. This table gives, for each state whose placement is not obvious from its statute, the reason for its group and the direction in which any misplacement would bias the estimates. Where a ban's legal status and access on the ground diverged, the analysis follows the statute in force for the rest of the window and notes the gap here. Sources as in Table A1.

Table A3—Source of the Pregnancy Start Date

Dating Source	**Ban States, %**	**Protected States, %**
Gestational-week diagnosis code (Z3A)	87.051	75.345
Ending-type default	9.112	19.579
First prenatal visit default	2.804	2.862
Embryo transfer	0.711	1.647
Cell-free DNA screening	0.175	0.205
Glucose tolerance test	0.063	0.115
Ending-type default, multiple birth	0.028	0.048
Maternal serum screening	0.023	0.068
Nuchal translucency ultrasound	0.021	0.102
Fetal anatomy ultrasound	0.012	0.029

Notes: Cells are the percent of pregnancy episodes in the continuous-enrollment sample dated by each source, under the priority order in Appendix A2; rows are sorted by the ban-state share.

Table A4—Sample Construction

Restriction	Continuous Enrollment	Episode Enrollment
All pregnancy episodes meeting the enrollment requirement	1,848,589	3,021,280
Ban or protected state at pregnancy start	1,576,819	2,495,494
Pregnancy start in 2020 or later*	1,135,994	2,495,494
Pregnancy start on or before February 2024 (40 weeks observable)	996,163	2,418,056
Exclude ended episodes with implausible gestation	995,377	2,416,491
Analysis sample	**995,377**	**2,416,491**
Ban states	262,367	798,269
Protected states	733,010	1,618,222
Event-study sample†	**818,245**	**2,000,319**
Ban states, in own cohort's window	155,502	485,714
Protected states	662,743	1,514,605

Notes: Rows count the pregnancy episodes remaining after each restriction in the continuous-enrollment and episode-enrolled samples (Section II.D and Appendix A4). *This step excludes no episode-enrolled pregnancy. That sample requires enrollment from twelve months before the pregnancy start, and the enrollment data begin in January 2019, so no pregnancy that started in 2019 can qualify; every episode-enrolled pregnancy already starts in 2020 or later. †These rows count the distinct pregnancies in each sample used by the event studies and pooled estimates (Section IV), the ban-state pregnancies conceived in their own cohort's eleven-quarter window (Table A5, quarters −6 to 4, excluding Indiana and South Carolina) and the protected-state pregnancies conceived from the second quarter of 2020 through the fourth quarter of 2023. Because a control pregnancy enters every sub-experiment whose window covers it, the stacked regression has more observations than these counts (1,607,569 in the continuous-enrollment sample for outcomes measured on all pregnancies).

Table A5—Pregnancies in Each Ban Cohort by Quarter Relative to the Ban

Quarter from Ban	Banned at *Dobbs*	Texas	West Virginia	Indiana and South Carolina	All Ban States
First treated quarter	2022Q3	2021Q4	2022Q4	2023Q4	
Pregnancies Conceived in the Quarter					
−6	7,376	6,484	241	1,555	15,656
−5	7,050	6,798	226	1,571	15,645
−4	6,810	7,517	238	1,625	16,190
−3	7,588	7,138	211	1,590	16,527
−2	7,050	7,176	253	1,643	16,122
−1	6,900	6,929	217	1,481	15,527
0	6,706	7,185	262	1,539	15,692
1	7,231	6,781	236	1,001	15,249
2	7,076	6,248	239	–	13,563
3	7,392	6,187	198	–	13,777
4	6,866	6,485	208	–	13,559
Total, quarters −6 to 4	**78,045**	**74,928**	**2,529**	**12,005**	**167,507**
Total, every quarter observed	**118,684**	**113,751**	**3,854**	**26,078**	**262,367**

Notes: Cells count ban-state pregnancies in the continuous-enrollment sample by quarter of conception relative to each cohort's first fully treated quarter, the series behind Figure 1. The *Dobbs* cohort is Alabama, Arkansas, Georgia, Idaho, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, South Dakota, and Tennessee. A dash marks a quarter whose pregnancies would be conceived after the data end, which is also why Indiana and South Carolina are not a sub-experiment of the stacked design.

Table A6—Pre-Period Balance Between Ban and Protected States

	Ban States Mean	Protected States Mean	Normalized Difference
Characteristics Fixed Before Pregnancy			
Age (years)	29.014	30.352	−0.21
Under 25	0.267	0.193	0.18
Race: White	0.416	0.459	−0.09
Race: Black	0.192	0.140	0.14
Race: Asian	0.019	0.039	−0.12
Race: unknown	0.334	0.215	0.27
Ethnicity: Hispanic	0.087	0.238	−0.42
Ethnicity: unknown	0.406	0.260	0.31
Medicaid	0.424	0.534	−0.22
Commercial insurance	0.575	0.464	0.22
First observed pregnancy	0.257	0.257	0.00
Multiple gestation	0.020	0.018	0.01
Obstetric comorbidity score	8.744	8.372	0.03
Number of comorbid conditions	0.776	0.719	0.05
Any comorbid condition	0.437	0.420	0.03
Pre-existing hypertension	0.067	0.051	0.07
Pre-existing obesity	0.169	0.180	−0.03
Pre-existing drug or alcohol use disorder	0.048	0.049	−0.01
Any of hypertension, obesity, or substance use	0.236	0.240	−0.01
Pre-existing pulmonary hypertension	0.000	0.001	−0.01
Maternal Health			
Any adverse pregnancy outcome	0.286	0.261	0.06
Severe maternal morbidity	0.028	0.027	0.01
Hypertensive disorder	0.213	0.157	0.14
Preterm birth (live births)	0.173	0.149	0.06
Timing and Intensity of Care			
Length of pregnancy, days	225.000	201.728	0.26
Days to first visit	45.992	45.261	0.02
Prolonged delivery stay (live births)	0.233	0.243	−0.02
Number of visits	16.203	16.316	−0.01
Pregnancy Endings			
Live birth	0.793	0.671	0.28
Miscarriage	0.157	0.142	0.04
Ending not observed	0.064	0.049	0.06
Insurance-billed abortion	0.012	0.152	−0.53

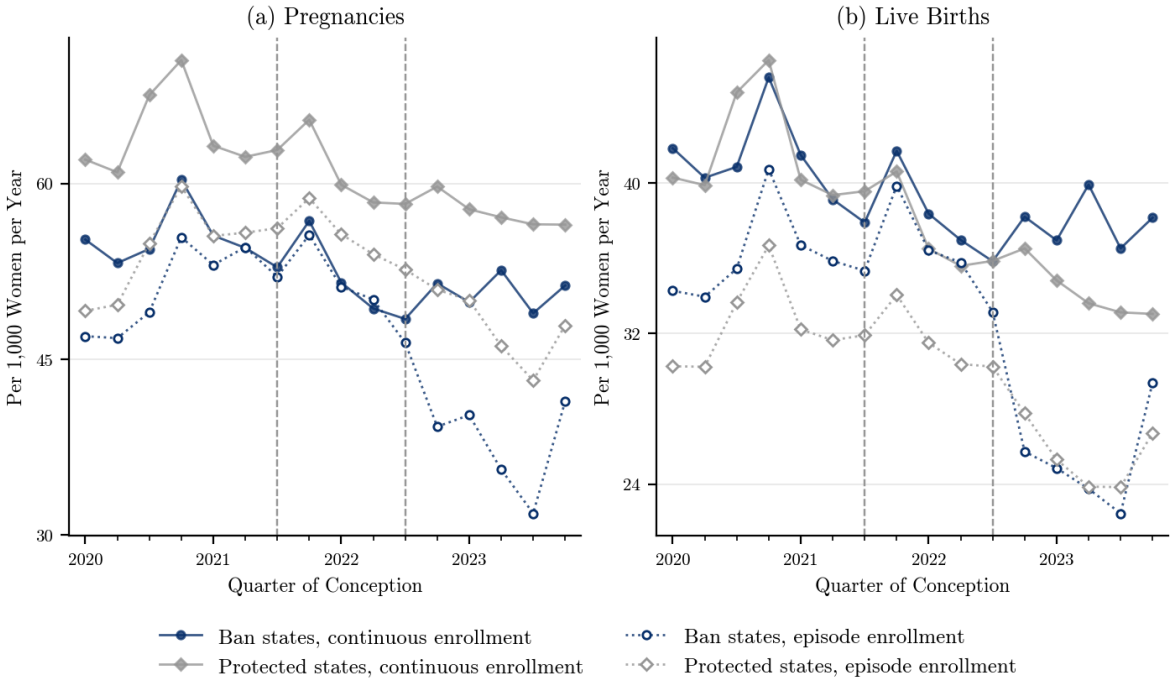
Notes: The table reports pre-ban means in the continuous-enrollment sample and normalized differences (the difference in means divided by the square root of the average of the two variances). Means are proportions except for age, the comorbidity score and count, the length of pregnancy and days to the first visit (in days), and the number of visits. Live birth, miscarriage, insurance-billed abortion, the length of pregnancy, and the days to the first visit are among pregnancies with an observed ending, preterm birth and a prolonged delivery stay among live births, and the other rows among all pregnancies. Normalized differences above 0.25 in absolute value, for Hispanic ethnicity, unknown race and ethnicity, the length of pregnancy, live birth, and insurance-billed abortion, are level differences that the state fixed effects absorb.

Table A7—The Continuous-Enrollment Sample and U.S. Births

	Continuous-Enrollment Sample	U.S. Births, 2023
Pregnancies, %		
Mean age of mother, years	29.9	29.6
Medicaid	50.7	41.5
Commercial or private insurance	49.1	51.0
Mother under 25	22.7	21.1
White	44.9	49.7
Black	15.1	13.7
Asian	3.3	6.0
Hispanic	21.3	26.3
Race unknown	24.4	–
Live Births, %		
Cesarean delivery	31.0	32.3
Gestational diabetes	11.4	8.3

Notes: Column 1 pools ban and protected states in the continuous-enrollment sample for pregnancies started January 2020 to February 2024, weighting by pregnancies; its upper rows are percentages of all pregnancies and its lower rows percentages of live births. Column 2 is every U.S. birth in 2023 from the final natality data (Osterman et al., 2025), with mean age from Brown et al. (2025); race and Hispanic origin refer to non-Hispanic single-race mothers and Hispanic mothers of any race, and payer is the source of payment for the delivery. Mean age is in years, and a dash marks a row with no counterpart in the natality data. Race is unknown for many pregnancies in the claims (the race unknown row), so the race rows are not strictly comparable, and birth certificates miss many complications that claims record (Bell, Anjur-Dietrich and Gemmill, 2026), so the gestational diabetes rows also differ in how they are measured.

Figure A1. Pregnancy and Birth Rates in the Two Enrollment Samples



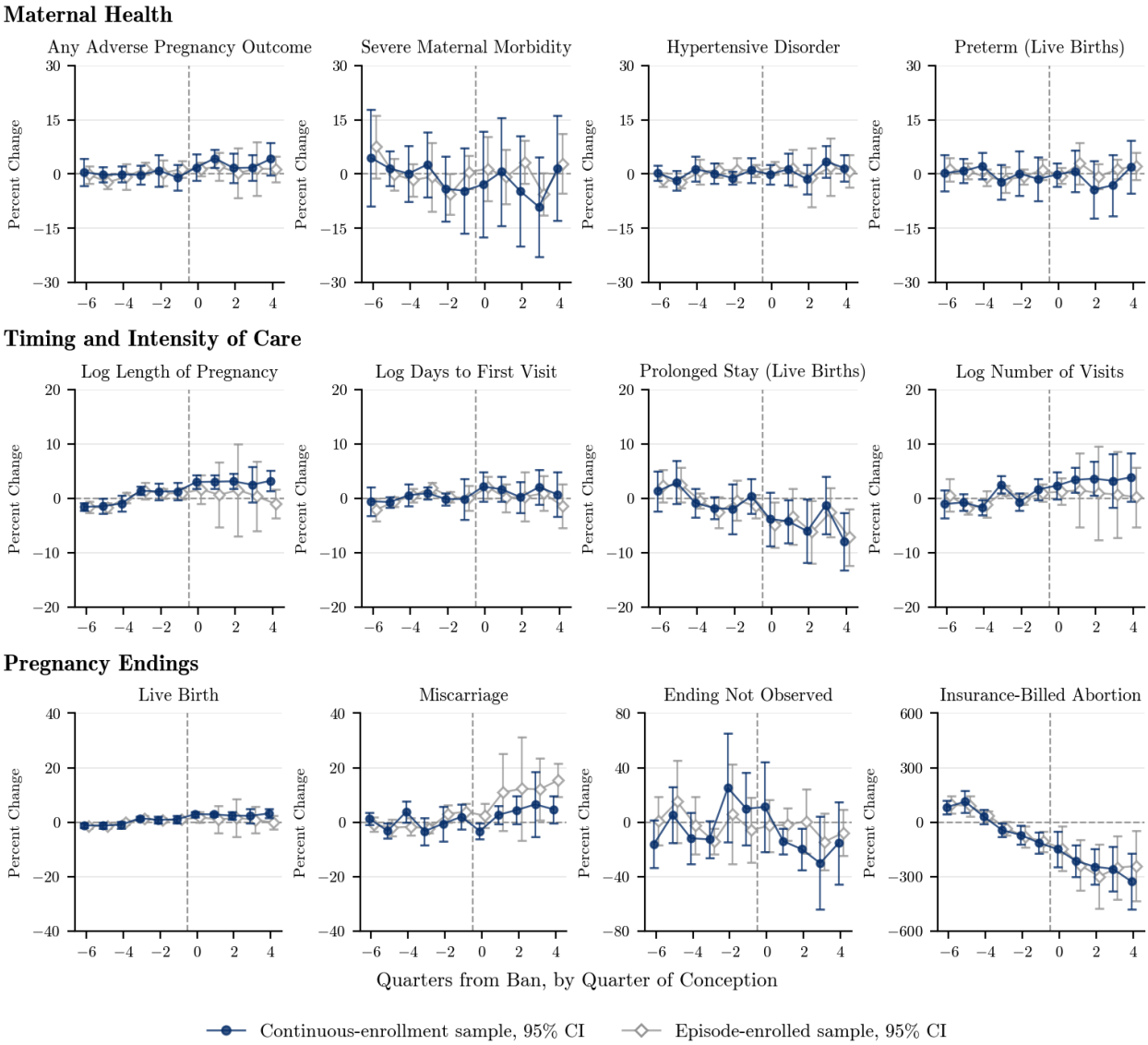
Notes: The figure plots pregnancies and live births per 1,000 enrolled women aged 15 to 45 per year, by quarter of conception, in ban and protected states, in the continuous-enrollment sample (solid lines, filled markers, as in Figure 4) and the episode-enrolled sample (dotted lines, hollow markers). Monthly rates are averaged to quarters. Ban states are circles and protected states diamonds. Dashed lines mark the third quarters of 2021 (Texas Senate Bill 8) and 2022 (*Dobbs*).

Table A8—Estimates by Enrollment Sample

Outcome	Episode-Enrolled Sample				Continuous-Enrollment Sample			
	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Pre-Ban Mean (5)	Estimate (SE) (6)	Percent of Mean (7)	95% CI, Percent of Mean (8)
Maternal Health								
Any adverse pregnancy outcome	0.278	0.004 (0.005)	1.3	[−2.4, 4.9]	0.285	0.008 (0.003)	2.8	[0.6, 4.9]
Severe maternal morbidity	0.027	0.000 (0.001)	0.5	[−5.4, 6.4]	0.026	−0.001 (0.001)	−2.9	[−11.4, 5.6]
Hypertensive disorder	0.202	0.002 (0.004)	0.9	[−3.5, 5.4]	0.208	0.002 (0.003)	1.0	[−1.5, 3.4]
Preterm birth (live births)	0.182	0.002 (0.002)	1.0	[−1.3, 3.4]	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]
Timing and Intensity of Care								
Log length of pregnancy	5.242	0.006 (0.023)	0.6	[−4.0, 5.3]	5.272	0.030 (0.006)	3.0	[1.8, 4.2]
Log days to first visit	3.676	0.004 (0.014)	0.4	[−2.3, 3.2]	3.616	0.014 (0.012)	1.4	[−1.1, 3.8]
Prolonged delivery stay (live births)	0.246	−0.011 (0.005)	−4.6	[−9.1, −0.1]	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]
Log number of visits	2.397	0.009 (0.025)	0.9	[−4.2, 6.1]	2.496	0.033 (0.014)	3.3	[0.4, 6.2]
Pregnancy Endings								
Live birth	0.772	0.008 (0.014)	1.0	[−2.8, 4.8]	0.789	0.022 (0.004)	2.7	[1.7, 3.8]
Miscarriage	0.170	0.017 (0.007)	10.1	[1.2, 19.0]	0.162	0.005 (0.003)	3.0	[−0.9, 6.9]
Ending not observed	0.097	−0.004 (0.005)	−4.4	[−15.1, 6.3]	0.062	−0.008 (0.005)	−13.3	[−28.3, 1.6]
Insurance-billed abortion	0.012	−0.028 (0.009)	−225.1	[−377.7, −72.5]	0.011	−0.025 (0.006)	−237.1	[−344.5, −129.6]

Notes: The table reports stacked difference-in-differences estimates in the Controls specification, with standard errors (SE), clustered by state, in parentheses. Columns 1 to 4 use the episode-enrolled sample (Appendix A4) and columns 5 to 8 the continuous-enrollment sample of Table 2. Columns 1 and 5 are each sample's ban-state mean in event quarters −6 to −1; columns 1, 2, 5, and 6 are in each outcome's units. Columns 3 and 7 are the estimate as a percent of that sample's pre-ban mean (100 times the coefficient for outcomes in logs), and columns 4 and 8 its 95 percent confidence interval (CI).

Figure A2. Event Studies, Continuous-Enrollment and Episode-Enrolled Samples



Notes: The figure plots stacked event-study coefficients in the continuous-enrollment sample (filled markers, as in Figure 2) and the episode-enrolled sample (hollow markers), in the Controls specification. Bars show 95 percent confidence intervals from standard errors clustered by state. The pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. The vertical axis, percent change, is each coefficient as a percent of its sample's ban-state pre-ban mean for shares and 100 times the coefficient for outcomes in logs, with axes drawn as in Figure 2.

Table A9—Subgroup Estimates for the Pregnancy Outcomes

Outcome	All Pregnancies (1)	Medicaid (2)	Minority (3)	High Risk (4)	Under 25 (5)	Excluding Texas (6)
Maternal Health						
Any adverse pregnancy outcome	2.8 (1.1)	4.2 (1.7)	2.9 (2.0)	4.0 (1.9)	6.3 (1.8)	3.4 (1.9)
	[0.6, 4.9]	[0.7, 7.7]	[−1.0, 6.9]	[0.1, 7.8]	[2.8, 9.9]	[−0.5, 7.3]
Pre-ban mean	0.285	0.322	0.313	0.315	0.286	0.295
Severe maternal morbidity	−2.9 (4.2)	−6.3 (8.0)	−7.5 (6.3)	−2.2 (4.6)	−6.6 (8.5)	−2.1 (6.8)
	[−11.4, 5.6]	[−22.6, 10.0]	[−20.2, 5.2]	[−11.4, 7.1]	[−23.7, 10.6]	[−15.9, 11.8]
Pre-ban mean	0.026	0.035	0.036	0.037	0.026	0.029
Hypertensive disorder	1.0 (1.2)	−1.6 (2.0)	0.7 (2.0)	0.1 (1.4)	2.0 (2.7)	−0.9 (1.6)
	[−1.5, 3.4]	[−5.7, 2.5]	[−3.3, 4.8]	[−2.8, 3.1]	[−3.5, 7.5]	[−4.1, 2.3]
Pre-ban mean	0.208	0.247	0.238	0.264	0.203	0.234
Preterm birth (live births)	−0.9 (1.8)	−2.3 (2.6)	−3.2 (3.3)	−1.7 (1.8)	0.0 (2.7)	−1.8 (2.6)
	[−4.5, 2.6]	[−7.5, 2.9]	[−9.8, 3.5]	[−5.4, 2.0]	[−5.6, 5.6]	[−7.1, 3.5]
Pre-ban mean	0.171	0.223	0.218	0.220	0.172	0.195
Timing and Intensity of Care						
Log length of pregnancy	3.0 (0.6)	3.1 (1.1)	4.1 (1.3)	3.0 (0.8)	3.9 (1.0)	2.5 (1.0)
	[1.8, 4.2]	[0.7, 5.4]	[1.5, 6.6]	[1.5, 4.5]	[1.9, 5.9]	[0.5, 4.5]
Pre-ban mean	5.272	5.255	5.223	5.234	5.315	5.262
Log days to first visit	1.4 (1.2)	−0.4 (1.7)	0.2 (1.2)	0.7 (1.4)	0.5 (1.3)	0.0 (1.8)
	[−1.1, 3.8]	[−3.9, 3.1]	[−2.2, 2.5]	[−2.1, 3.5]	[−2.2, 3.2]	[−3.5, 3.6]
Pre-ban mean	3.616	3.616	3.643	3.490	3.635	3.604
Prolonged delivery stay (live births)	−4.6 (2.0)	−2.4 (3.2)	−5.6 (2.6)	−4.0 (2.9)	−4.6 (2.6)	−1.8 (2.6)
	[−8.6, −0.6]	[−8.8, 4.1]	[−10.8, −0.4]	[−9.9, 1.9]	[−9.9, 0.6]	[−7.1, 3.5]
Pre-ban mean	0.233	0.261	0.265	0.246	0.303	0.247
Log number of visits	3.3 (1.4)	4.1 (2.1)	4.3 (1.8)	3.1 (1.6)	5.1 (1.9)	3.8 (1.8)
	[0.4, 6.2]	[−0.1, 8.3]	[0.7, 7.8]	[−0.1, 6.4]	[1.3, 8.9]	[0.1, 7.5]
Pre-ban mean	2.496	2.566	2.465	2.621	2.535	2.505
Pregnancy Endings						
Live birth	2.7 (0.5)	3.0 (0.9)	3.8 (1.1)	2.6 (0.6)	3.6 (0.9)	2.3 (0.8)
	[1.7, 3.8]	[1.2, 4.8]	[1.6, 6.0]	[1.4, 3.8]	[1.8, 5.5]	[0.7, 3.8]
Pre-ban mean	0.789	0.785	0.763	0.771	0.819	0.787
Miscarriage	3.0 (1.9)	0.5 (4.2)	4.3 (4.0)	2.2 (1.9)	2.9 (3.6)	3.1 (3.4)
	[−0.9, 6.9]	[−8.1, 9.0]	[−3.9, 12.5]	[−1.6, 6.0]	[−4.4, 10.1]	[−3.8, 10.1]
Pre-ban mean	0.162	0.155	0.171	0.172	0.133	0.157
Ending not observed	−13.3 (7.4)	−25.6 (16.9)	−16.7 (9.4)	−15.6 (9.0)	−22.6 (16.2)	−20.1 (12.2)
	[−28.3, 1.6]	[−59.8, 8.7]	[−35.7, 2.4]	[−33.8, 2.6]	[−55.4, 10.2]	[−44.9, 4.7]
Pre-ban mean	0.062	0.067	0.071	0.076	0.055	0.067
Insurance-billed abortion	−237.1 (53.0)	−192.5 (63.1)	−237.8 (50.7)	−179.4 (43.2)	−285.1 (75.3)	−176.8 (47.9)
	[−344.5, −129.6]	[−320.5, −64.5]	[−340.7, −134.8]	[−267.0, −91.9]	[−437.7, −132.6]	[−274.0, −79.6]
Pre-ban mean	0.011	0.011	0.015	0.011	0.012	0.012

Notes: Cells are stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, each as a percent of the subgroup's pre-ban mean (100 times the coefficient for outcomes in logs), with the standard error, clustered by state, in parentheses, the 95 percent confidence interval in brackets, and the group's ban-state mean in event quarters −6 to −1, in the outcome's units, beneath. The All Pregnancies column repeats Table 2, and the subgroups are defined as in Figure 3.

Table A10—Maternal Health and Care within Live Births and within Losses

Outcome	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)
A. Live Births				
Any adverse pregnancy outcome	0.369	−0.005 (0.003)	−1.2	[−3.1, 0.6]
Severe maternal morbidity	0.030	−0.003 (0.001)	−9.5	[−17.6, −1.3]
Hypertensive disorder	0.262	−0.008 (0.003)	−3.0	[−5.7, −0.3]
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]
Log length of pregnancy	5.579	−0.001 (0.001)	−0.1	[−0.4, 0.1]
Log days to first visit	3.653	0.013 (0.013)	1.3	[−1.4, 4.0]
Prolonged delivery stay (live births)	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]
Log number of visits	2.864	−0.020 (0.011)	−2.0	[−4.3, 0.3]
B. Losses				
Any adverse pregnancy outcome	0.022	0.001 (0.002)	5.5	[−10.2, 21.2]
Severe maternal morbidity	0.014	0.002 (0.002)	11.3	[−10.7, 33.4]
Hypertensive disorder	0.035	−0.003 (0.001)	−8.6	[−16.0, −1.3]
Log length of pregnancy	4.113	0.010 (0.005)	1.0	[0.1, 2.0]
Log days to first visit	3.464	0.026 (0.012)	2.6	[0.0, 5.1]
Log number of visits	1.338	−0.013 (0.013)	−1.3	[−3.9, 1.4]

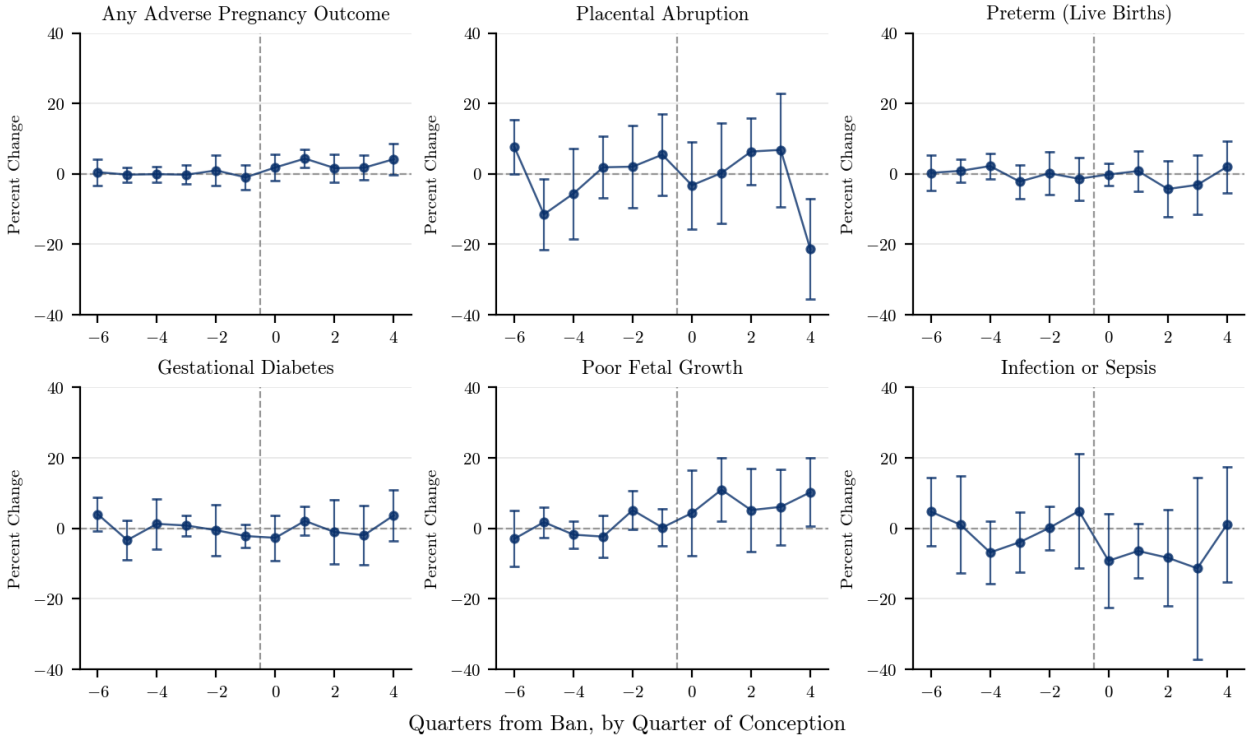
Notes: The table reports stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, among pregnancies that ended in a live birth (Panel A) and among pregnancies that ended in a miscarriage, stillbirth, or ectopic pregnancy (Panel B). Column 1 is the group's ban-state mean in event quarters −6 to −1 and column 2 the estimate, both in each outcome's units, with its standard error (SE), clustered by state, in parentheses. Column 3 is the estimate as a percent of that mean (100 times the coefficient for outcomes in logs) and column 4 its 95 percent confidence interval (CI). Preterm birth and a prolonged delivery stay are defined only for live births, and the ending outcomes are not estimated, because each group fixes the ending. Because the bans changed how pregnancies ended, these estimates condition on an outcome of the bans, so they describe how the pooled estimates divide across endings rather than identify effects within each.

Table A11—The Adverse-Outcome Composite and Its Components

		Controls			Raw	
Outcome	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Estimate (SE) (5)	Percent of Mean (6)
Composite						
Any adverse pregnancy outcome	0.2847	0.0079 (0.0030)	2.8	[0.6, 4.9]	0.0090 (0.0032)	3.2
Components						
Placental abruption	0.0117	−0.0003 (0.0004)	−2.2	[−9.0, 4.7]	−0.0002 (0.0004)	−2.0
Preterm birth (live births)	0.1713	−0.0016 (0.0030)	−0.9	[−4.5, 2.6]	−0.0013 (0.0031)	−0.8
Gestational diabetes	0.0937	0.0000 (0.0024)	0.0	[−5.2, 5.2]	−0.0012 (0.0020)	−1.3
Poor fetal growth	0.1015	0.0075 (0.0047)	7.4	[−2.1, 16.8]	0.0089 (0.0049)	8.7
Infection or sepsis	0.0240	−0.0016 (0.0012)	−6.8	[−17.0, 3.4]	−0.0011 (0.0012)	−4.6

Notes: The table reports stacked difference-in-differences estimates in the continuous-enrollment sample, laid out as Table 2 without its pre-trend test, with the same fixed effects and weights and with standard errors (SE), clustered by state, in parentheses. The composite equals one if a pregnancy has any of the five components below it. Preterm birth is measured among live births. Poor fetal growth is diagnosed only in pregnancies that continue, and in one validation study about a third of claims diagnoses of a small-for-gestational-age infant were confirmed in medical records (Chomistek et al., 2023).

Figure A3. Event Studies for the Adverse-Outcome Composite and Its Components



Notes: The figure plots stacked difference-in-differences event-study coefficients by quarter of conception relative to the ban, in the Controls specification and the continuous-enrollment sample, for the composite (first panel, as in Figure 2) and its five components. Bars show 95 percent confidence intervals from standard errors clustered by state. The vertical axis, percent change, is each coefficient in percent of the outcome's ban-state pre-ban mean, the pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. Axes are as in Figure 2.

Table A12—Fertility: Pregnancy and Live-Birth Rates among Enrolled Women

Group	Pre-Ban Mean (1)	State Fixed Effects			Raw	
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Estimate (SE) (5)	Percent of Mean (6)
Pregnancy Rate						
All	52.09	0.11 (0.61)	0.2	[−2.2, 2.6]	−0.31 (0.74)	−0.6
Medicaid	67.82	0.11 (0.98)	0.2	[−2.8, 3.1]	−1.63 (1.86)	−2.4
Minority	63.12	−2.25 (0.65)	−3.6	[−5.6, −1.5]	−3.54 (1.04)	−5.6
High risk[a]	53.17	0.85 (0.66)	1.6	[−0.9, 4.1]	0.10 (1.00)	0.2
Under 25	41.06	1.16 (1.59)	2.8	[−5.0, 10.6]	0.04 (1.27)	0.1
Excluding Texas	52.30	0.39 (0.63)	0.8	[−1.7, 3.2]	0.57 (1.04)	1.1
Live-Birth Rate						
All	38.50	1.41 (0.57)	3.7	[0.7, 6.7]	1.11 (0.71)	2.9
Medicaid	49.04	2.28 (1.03)	4.7	[0.4, 8.9]	0.81 (1.68)	1.7
Minority	43.73	0.87 (0.87)	2.0	[−2.0, 6.0]	−0.09 (1.17)	−0.2
High risk[a]	38.11	2.10 (0.62)	5.5	[2.2, 8.8]	1.45 (0.87)	3.8
Under 25	31.57	2.12 (1.20)	6.7	[−1.0, 14.4]	1.22 (0.98)	3.9
Excluding Texas	38.62	1.56 (0.74)	4.0	[0.2, 7.9]	1.70 (0.93)	4.4

Notes: The table reports stacked difference-in-differences (DiD) estimates on the panel of enrolled women aged 15 to 45 by state and quarter, weighted by enrolled women, with standard errors (SE), clustered by state, in parentheses. State Fixed Effects has event-time and sub-experiment by state fixed effects; Raw has event-time fixed effects and a treated-state indicator. Rates are pregnancies or live births per 1,000 enrolled women per year (conceptions in the state and quarter divided by woman-months of enrollment, times 12,000), so they do not condition on who becomes pregnant. Column 1 is the mean in treated states before the ban; columns 1, 2, and 5 are in rates. Column 3 is the estimate as a percent of that mean, and column 4 its 95 percent confidence interval (CI). Unlike the pregnancy estimates, this stack keeps Indiana and South Carolina as a fourth sub-experiment, so it has 40 clusters. Because a pregnancy that started after February 2024 cannot be followed to its ending, the rates stop with the fourth quarter of 2023, so this sub-experiment adds its pre-ban quarters and only its first post-ban quarter. Each estimate compares the post-ban quarters with the average of quarters −6 to −1. Minority women have a Black race code or a Hispanic ethnicity code. [a]High-risk women have any condition of the obstetric comorbidity index in the preceding 12 months.

Table A13—Composition of the Pregnant Population

Characteristic	Pre-Ban Mean (1)	Raw		
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)
Obstetric comorbidity score	8.387	0.033 (0.139)	0.4	[−3.0, 3.7]
Number of obstetric comorbidities	0.741	0.005 (0.012)	0.7	[−2.5, 3.9]
Any obstetric comorbidity	0.423	0.002 (0.005)	0.4	[−1.8, 2.6]
Medicaid	0.355	0.008 (0.007)	2.2	[−1.9, 6.4]
Under age 25	0.247	0.028 (0.009)	11.2	[3.8, 18.5]

Notes: The table reports stacked difference-in-differences (DiD) estimates in the continuous-enrollment sample for pre-pregnancy health measures and the Medicaid and under-25 shares of pregnancies, with standard errors (SE), clustered by state, in parentheses. The columns are as under Controls in Table 2 without the pre-trend test, but the estimates use the Raw specification because these characteristics are themselves controls. A ban cannot change these measures for any one woman, so a nonzero estimate means that the set of women who became pregnant changed. The comorbidity score weights conditions by the points of Leonard et al. (2020), as in the regression controls and Table 1.

Table A14—Sub-Experiments in the Stacked Design

First Treated Quarter	Treated States	Treated Pregnancies	Control Pregnancies	Share of Treated Pregnancies, %
2021Q4	TX	74,928	493,967	48.2
2022Q3	AL, AR, GA, ID, KY, LA, MO, MS, OK, SD, TN	78,045	480,762	50.2
2022Q4	WV	2,529	477,338	1.6

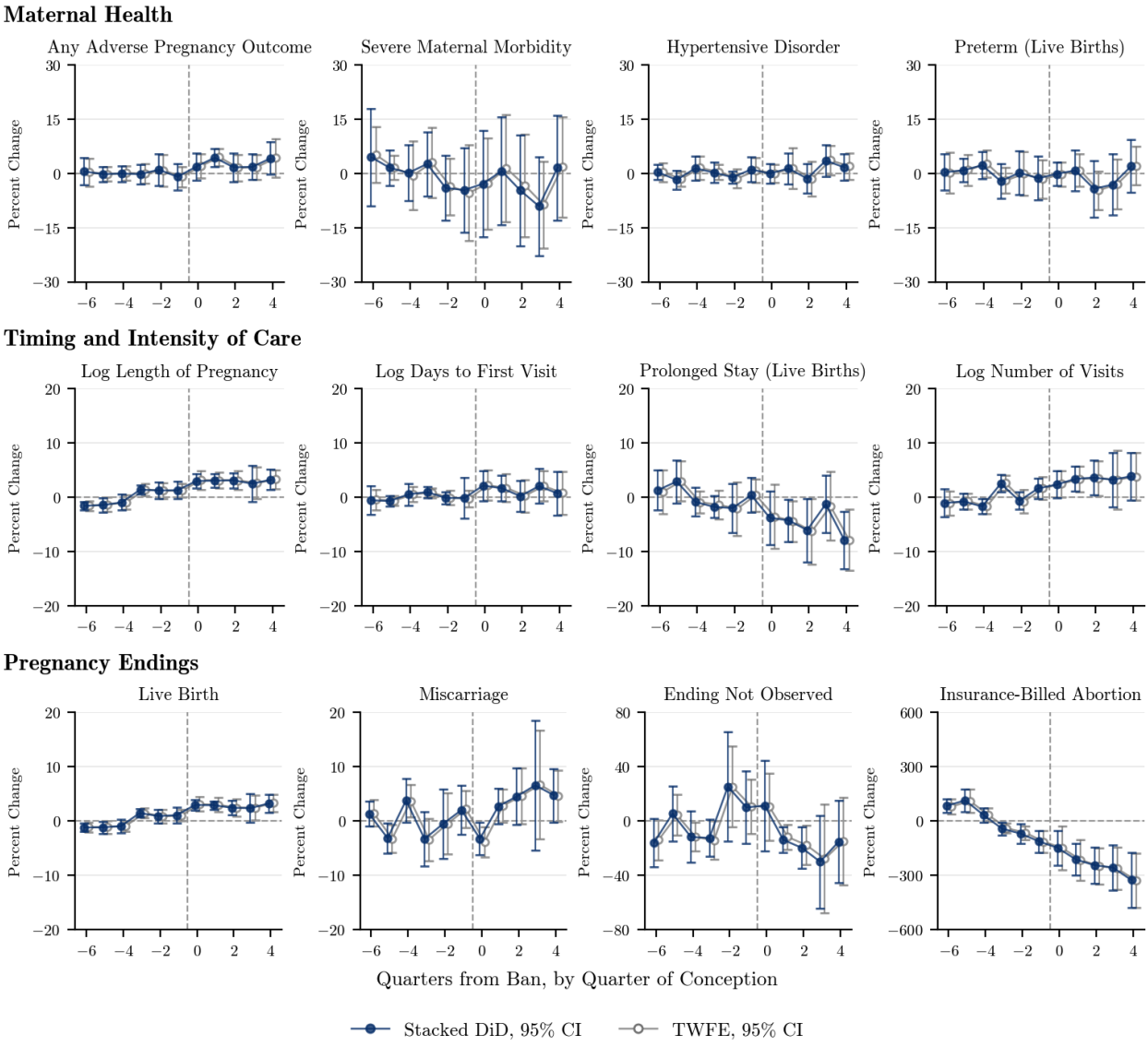
Notes: Each row is a ban cohort, defined by its first fully treated quarter, in the continuous-enrollment sample. Each sub-experiment pairs the cohort with the protected states over its event window, quarters −6 to 4. Treated pregnancies are the cohort's pregnancies conceived in the window, and control pregnancies are the protected-state pregnancies conceived in it, so a protected-state pregnancy is counted in every sub-experiment whose window covers it. The last column is each cohort's share of the treated pregnancies in the stacked design. The window of Indiana and South Carolina runs past the data, so they are not a sub-experiment (Table A5).

Table A15—Stacked Estimates by Reference Period

		Pre-Ban Average		Quarter −2		
Outcome	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	Estimate (SE) (4)	Percent of Mean (5)	95% CI, Percent of Mean (6)
Maternal Health						
Any adverse pregnancy outcome	0.285	0.008 (0.003)	2.8	0.005 (0.006)	1.8	[−2.4, 5.9]
Severe maternal morbidity	0.026	−0.001 (0.001)	−2.9	0.000 (0.001)	1.2	[−9.7, 12.1]
Hypertensive disorder	0.208	0.002 (0.003)	1.0	0.004 (0.002)	2.1	[−0.1, 4.2]
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	−0.002 (0.003)	−1.1	[−5.1, 2.9]
Timing and Intensity of Care						
Log length of pregnancy	5.272	0.030 (0.006)	3.0	0.017 (0.006)	1.7	[0.5, 3.0]
Log days to first visit	3.616	0.014 (0.012)	1.4	0.015 (0.016)	1.5	[−1.7, 4.7]
Prolonged delivery stay (live births)	0.233	−0.011 (0.005)	−4.6	−0.006 (0.005)	−2.6	[−6.8, 1.6]
Log number of visits	2.496	0.033 (0.014)	3.3	0.040 (0.011)	4.0	[1.7, 6.2]
Pregnancy Endings						
Live birth	0.789	0.022 (0.004)	2.7	0.015 (0.004)	1.9	[0.8, 3.0]
Miscarriage	0.162	0.005 (0.003)	3.0	0.006 (0.005)	3.6	[−2.9, 10.0]
Ending not observed	0.062	−0.008 (0.005)	−13.3	−0.024 (0.012)	−38.6	[−76.5, −0.7]
Insurance-billed abortion	0.011	−0.025 (0.006)	−237.1	−0.018 (0.005)	−166.9	[−271.3, −62.5]

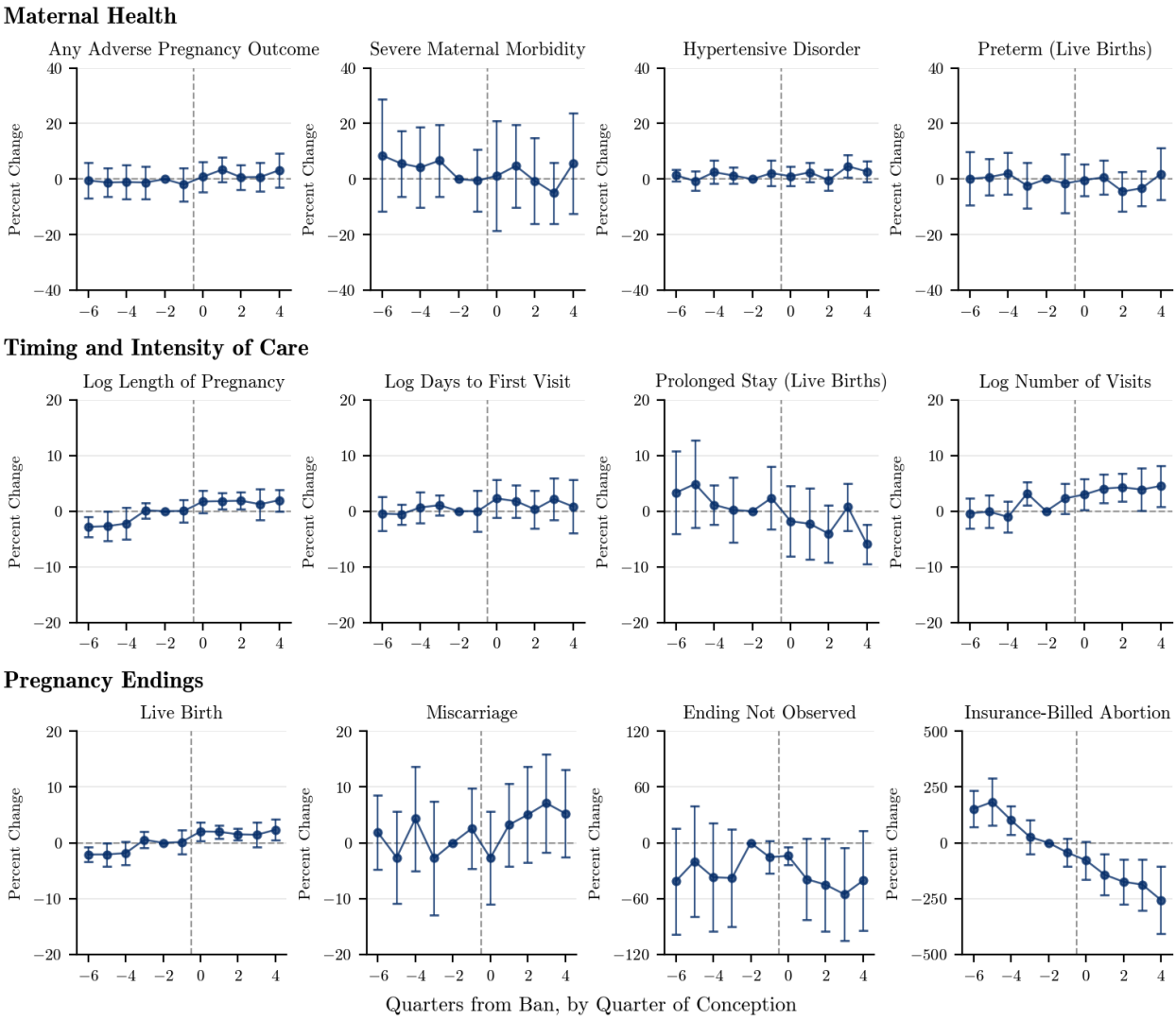
Notes: The table reports stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, laid out as Table 2 without its pre-trend test, with standard errors (SE), clustered by state, in parentheses. Columns 2 and 3 repeat columns 2 and 3 of Table 2, which compare the post-ban quarters with the average of quarters −6 to −1, and columns 4 to 6 compare them with quarter −2 alone, with the 95 percent confidence interval (CI) in column 6. The post-ban quarters are the same in both.

Figure A4. Event Studies, Stacked and Two-Way Fixed Effects



Notes: The figure plots stacked difference-in-differences event-study coefficients from Figure 2 (filled markers) and two-way fixed effects (TWFE) coefficients for the same outcome (hollow markers), in the Controls specification and the continuous-enrollment sample. Bars show 95 percent confidence intervals from standard errors clustered by state. The TWFE regression uses the pregnancies of the stacked design, with state and quarter-of-conception fixed effects and the Controls covariates, each pregnancy entering once and unweighted. In both, the pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. The vertical axis, percent change, is each coefficient as a percent of the ban-state pre-ban mean for shares and 100 times the coefficient for outcomes in logs, and vertical axes are drawn as in Figure 2.

Figure A5. Event Studies Against Quarter −2



Notes: The figure plots stacked difference-in-differences event-study coefficients by quarter of conception relative to the ban, quarters −6 to 4, estimated against quarter −2 alone, in the specification and sample of Table 2, the Controls specification and the continuous-enrollment sample. The quarter −2 coefficient is normalized to zero, and the dashed vertical line falls between quarters −1 and 0. Bars show 95 percent confidence intervals from standard errors clustered by state. The vertical axis, percent change, is each coefficient as a percent of the ban-state pre-ban mean for shares and 100 times the coefficient for outcomes in logs, and vertical axes are drawn as in Figure 2. Table A15 reports the pooled estimates against quarter −2.

Table A16—Stacked and Synthetic Difference-in-Differences Estimates

Outcome	Pre-Ban Mean (1)	Synthetic DiD, Controls: Estimate (SE) (2)	Synthetic DiD, Controls: Percent of Mean (3)	Synthetic DiD, Controls: 95% CI, Percent of Mean (4)	Synthetic DiD, Raw: Estimate (SE) (5)	Synthetic DiD, Raw: Percent of Mean (6)	Stacked DiD, Controls: Percent of Mean (7)
Maternal Health							
Any adverse pregnancy outcome	0.285	0.006 (0.014)	2.0	[−7.7, 11.7]	0.007 (0.014)	2.4	2.8
Severe maternal morbidity	0.026	−0.001 (0.004)	−4.3	[−32.6, 24.0]	−0.001 (0.004)	−4.1	−2.9
Hypertensive disorder	0.208	0.001 (0.011)	0.3	[−9.9, 10.6]	0.002 (0.011)	1.0	1.0
Preterm birth (live births)	0.171	−0.012 (0.013)	−7.1	[−21.5, 7.4]	−0.008 (0.011)	−4.9	−0.9
Timing and Intensity of Care							
Log length of pregnancy	5.272	0.009 (0.019)	0.9	[−2.9, 4.7]	0.011 (0.020)	1.1	3.0
Log days to first visit	3.616	0.011 (0.026)	1.1	[−4.0, 6.2]	0.011 (0.026)	1.1	1.4
Prolonged delivery stay (live births)	0.233	−0.019 (0.022)	−8.0	[−26.5, 10.6]	−0.017 (0.023)	−7.2	−4.6
Log number of visits	2.496	0.011 (0.034)	1.1	[−5.5, 7.6]	0.002 (0.032)	0.2	3.3
Pregnancy Endings							
Live birth	0.789	0.005 (0.013)	0.6	[−2.5, 3.7]	0.010 (0.015)	1.2	2.7
Miscarriage	0.162	0.021 (0.013)	13.1	[−2.4, 28.7]	0.017 (0.012)	10.2	3.0
Ending not observed	0.062	−0.009 (0.018)	−13.9	[−70.0, 42.1]	−0.008 (0.018)	−12.5	−13.3
Insurance-billed abortion	0.011	−0.023 (0.009)	−213.9	[−382.4, −45.4]	−0.023 (0.009)	−217.4	−237.1

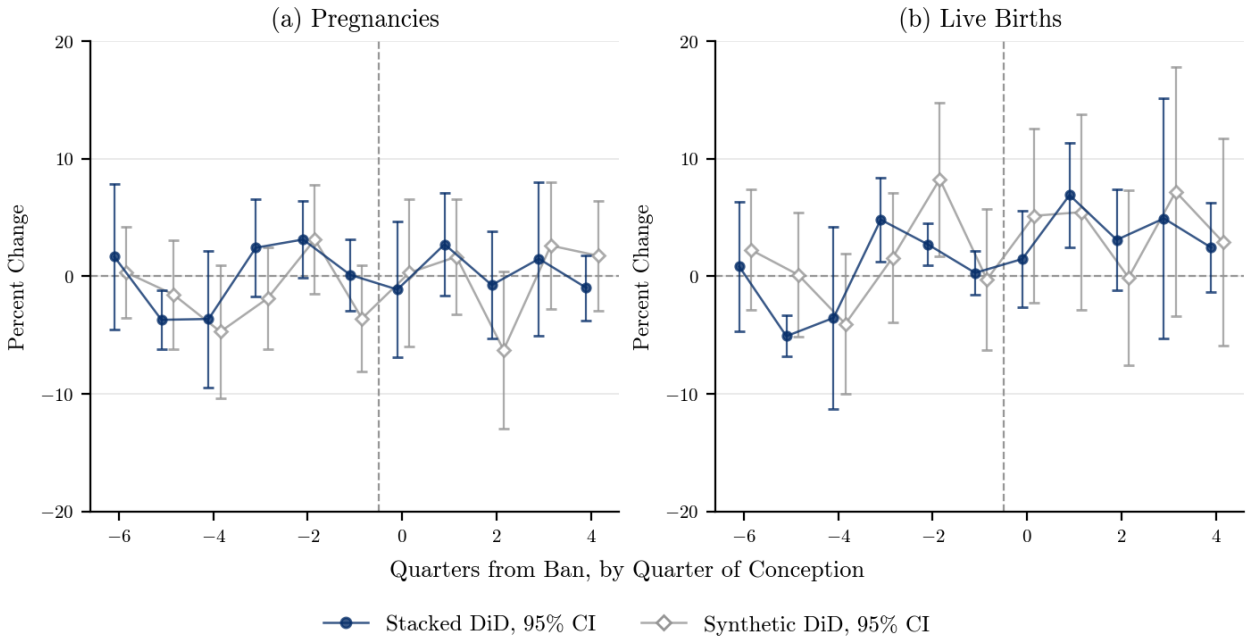
Notes: Column 1 is the ban-state mean in event quarters −6 to −1 of the stacked design, which scales every percent (100 times the coefficient for outcomes in logs). Columns 2 to 6 are synthetic difference-in-differences (DiD) estimates (Arkhangelsky et al., 2021) on state by quarter-of-conception means in the continuous-enrollment sample, conceptions from the first quarter of 2020 to the fourth quarter of 2023, excluding Indiana and South Carolina. Columns 2 to 4 average the outcome net of the Controls covariates, fitted on pregnancies with state and quarter fixed effects, and columns 5 and 6 the raw outcome. Standard errors (SE), in parentheses, come from 200 placebo replications (seed 20261013), and the 95 percent confidence interval (CI) in column 4 uses the normal critical value. Synthetic DiD has no pre-trend test. Column 7 repeats column 3 of Table 2.

Table A17—Fertility Estimates by Estimator

Group	Pre-Ban Mean (1)	Stacked DiD		Synthetic DiD	
		Percent of Mean (2)	95% CI, Percent of Mean (3)	Percent of Mean (4)	95% CI, Percent of Mean (5)
Pregnancy Rate					
All	52.09	0.2	[−2.2, 2.6]	1.9	[−3.7, 7.5]
Medicaid	67.82	0.2	[−2.8, 3.1]	−6.0	[−30.7, 18.8]
Minority	63.12	−3.6	[−5.6, −1.5]	−18.3	[−25.8, −10.8]
High risk[a]	53.17	1.6	[−0.9, 4.1]	4.4	[−2.8, 11.5]
Under 25	41.06	2.8	[−5.0, 10.6]	3.4	[−6.9, 13.8]
Excluding Texas	52.30	0.8	[−1.7, 3.2]	1.9	[−3.5, 7.3]
Live-Birth Rate					
All	38.50	3.7	[0.7, 6.7]	6.2	[−2.8, 15.2]
Medicaid	49.04	4.7	[0.4, 8.9]	1.9	[−36.1, 39.9]
Minority	43.73	2.0	[−2.0, 6.0]	−10.3	[−27.4, 6.8]
High risk[a]	38.11	5.5	[2.2, 8.8]	4.8	[−5.7, 15.4]
Under 25	31.57	6.7	[−1.0, 14.4]	4.5	[−7.2, 16.3]
Excluding Texas	38.62	4.0	[0.2, 7.9]	5.9	[−3.5, 15.3]

Notes: Estimates are for pregnancy and live-birth rates per 1,000 enrolled women aged 15 to 45 per year in the continuous-enrollment sample, as in Table A12. Column 1 is the mean in treated states before the ban in the stacked design, which scales every percent and 95 percent confidence interval (CI). Columns 2 and 3 repeat columns 3 and 4 of Table A12, which compare the post-ban quarters with the average of quarters −6 to −1. Columns 4 and 5 are synthetic difference-in-differences (DiD) estimates (Arkhangelsky et al., 2021) on state by quarter rates, each state unweighted, from the first quarter of 2020 through the fourth quarter of 2023, excluding Indiana and South Carolina, with standard errors from 200 placebo replications and the normal critical value. [a]High-risk women have any condition of the obstetric comorbidity index in the preceding 12 months.

Figure A6. Fertility Event Studies, Stacked and Synthetic Difference-in-Differences



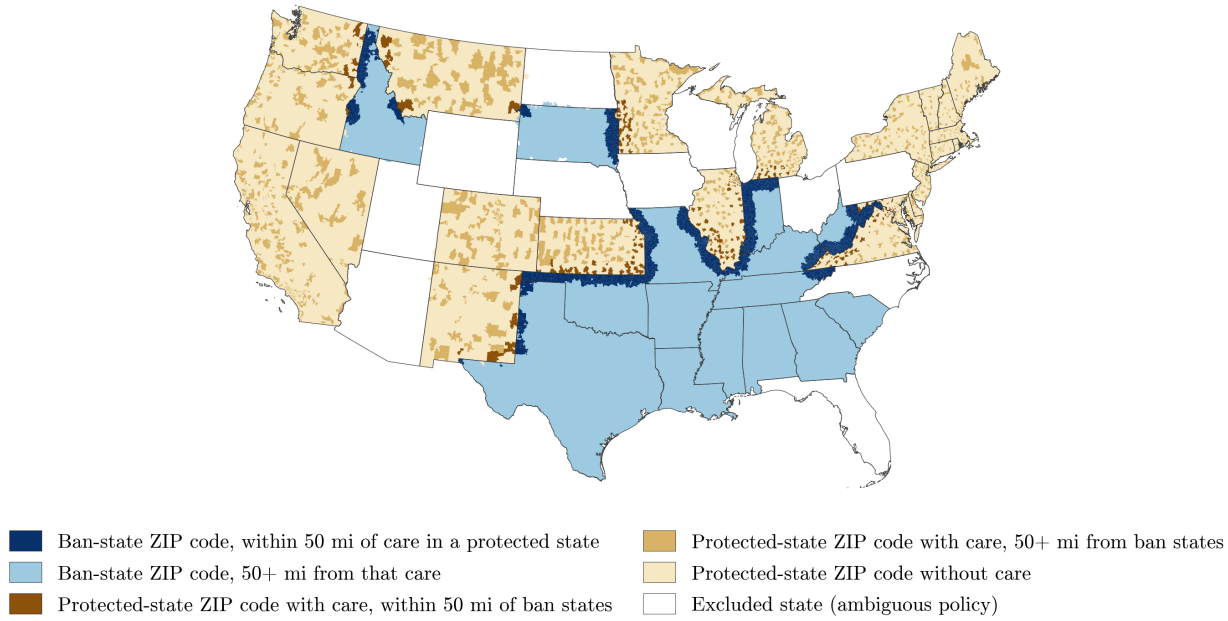
Notes: The figure plots coefficients for the pregnancy and live-birth rates of enrolled women aged 15 to 45, for all women. The vertical axis, percent change, is each coefficient as a percent of the stacked design's pre-ban rate in treated states, the mean that scales Table A17. Filled markers repeat panels (a) and (b) of Figure 4, whose pre-ban coefficients are normalized to average zero over quarters −6 to −1. Hollow markers are synthetic difference-in-differences coefficients on the state by quarter rates of Table A17, each the gap in that quarter between treated states and their synthetic control less the pre-ban gap under the synthetic time weights, averaged over ban cohorts by treated states, so their pre-ban coefficients need not average zero. Bars show 95 percent confidence intervals; for the hollow markers they use standard errors from 200 bootstrap draws of states, while the pooled estimates in Table A17 use placebo standard errors. The dashed vertical line falls between quarters −1 and 0, and vertical axes are drawn as in Figure 4.

Table A18—ZIP-Code Placement Diagnostics

	Within 50 Miles	**Beyond 50 Miles, by Ban Cohort**		
	of Protected Care	**Texas**	***Dobbs* Cohort**	**Late Adopters**
Visits in the year before pregnancy	21.09	17.26	22.11	20.01
in the modal ZIP code	7.20	5.94	7.03	6.88
in the second most common ZIP code	3.04	2.64	3.01	2.91
in the modal three-digit ZIP area	9.02	8.10	9.42	9.11
with no provider ZIP code	7.54	5.81	8.42	7.11
Unique ZIP codes visited	4.41	4.21	4.66	4.25
Share of visits with a provider ZIP in the modal ZIP code	**0.53**	**0.52**	**0.51**	**0.53**
in the second most common ZIP code	0.22	0.23	0.22	0.23
in the modal three-digit ZIP area	0.67	0.71	0.69	0.71
Pregnancies	28,285	95,685	94,265	17,070

Notes: Cells are means across women in the year before their first observed pregnancy, in the continuous-enrollment sample. Column 1 is the close-to-care sample, pregnancies whose assigned ban-state ZIP code lies within 50 miles of protected-state emergency care (Section A9); the other columns are assigned ZIP codes farther away, by ban cohort, with the late adopters West Virginia, Indiana, and South Carolina. A visit is located by the NPPES practice ZIP code of its rendering NPI, and the modal three-digit ZIP area counts visits to any ZIP code with the same first three digits.

Figure A7. Close-to-Care Classification of Lower-48 ZIP Codes



Notes: Each polygon is a 2020 Census ZIP Code Tabulation Area (ZCTA), with states classified as in Table A1 and excluded states in white. In ban states, ZCTAs within 50 miles of a ZCTA with observed emergency care (a general acute care hospital or emergency medicine provider) in a protected state are navy and the rest light blue, the close-to-care classification, drawn here from each ZCTA's interior point rather than the centroid the estimation uses. In protected states, ZCTAs with observed care within 50 miles of a ban-state ZCTA are dark brown, those with care farther away are tan, and those without observed care are cream. Distances use the Conus Albers equal-area projection (EPSG 5070). As the data-use terms require, the map shows every eligible ZIP code, not the ZIP codes where pregnancies are observed.

Table A19—Close-to-Care Sample Characteristics

	Beyond 50 Miles		Within 50 Miles	
	Before	**After**	**Before**	**After**
Pregnancies	107,453	99,567	18,678	9,607
Demographics, % Unless Noted				
Age, mean (SD)	29.10	28.99	28.30	27.85
	(6.40)	(6.83)	(6.41)	(6.71)
Under 25	26.1	30.0	31.4	36.6
Medicaid	42.2	36.1	53.6	55.2
Commercial	57.7	63.8	46.3	44.8
White	41.6	41.1	52.6	57.6
Black	20.0	16.5	14.9	11.6
Asian	1.9	1.9	0.9	0.8
Hispanic	9.3	14.4	9.3	12.8
First observed pregnancy	25.7	26.8	25.5	29.5
Health Before Pregnancy, % Unless Noted				
Hypertension, obesity, or substance use	25.8	27.4	28.0	33.2
Obstetric comorbidity score, mean (SD)	9.49	9.84	10.63	11.85
	(14.68)	(15.14)	(15.68)	(16.42)
Maternal Health, %				
Any adverse pregnancy outcome	28.7	29.2	28.4	28.8
Adjusted for characteristics	[28.7]	[29.3]	[28.2]	[28.2]
Severe maternal morbidity	2.8	2.6	2.9	3.0
Adjusted for characteristics	[2.9]	[2.6]	[2.8]	[2.8]
Hypertensive disorder	21.6	22.1	21.1	23.6
Adjusted for characteristics	[21.5]	[22.3]	[20.9]	[23.1]
Preterm birth (live births)	17.7	17.6	17.7	19.2
Adjusted for characteristics	[17.7]	[17.8]	[17.5]	[18.9]
Timing and Intensity of Care, % Unless Noted				
Length of pregnancy, days, mean (SD)	223.9	223.2	221.4	222.7
	(84.2)	(84.8)	(86.2)	(84.9)
Adjusted for characteristics	[223.9]	[223.7]	[220.2]	[219.8]
Days to first visit, mean (SD)	43.2	43.0	44.1	42.9
	(29.4)	(29.1)	(32.0)	(30.0)
Adjusted for characteristics	[43.3]	[43.1]	[43.6]	[42.5]
Prolonged delivery stay (live births)	23.5	24.1	23.2	23.5
Adjusted for characteristics	[23.7]	[23.9]	[23.4]	[22.9]
Number of visits, mean (SD)	16.6	16.7	17.1	18.0
	(11.0)	(10.9)	(11.6)	(11.8)
Adjusted for characteristics	[16.6]	[16.8]	[16.8]	[17.3]
Pregnancy Endings, % of Observed Endings Unless Noted				
Live birth	78.8	78.5	77.5	78.1
Adjusted for characteristics	[78.8]	[78.8]	[76.8]	[76.7]
Miscarriage	16.0	16.6	15.5	15.4
Adjusted for characteristics	[16.1]	[16.3]	[16.1]	[16.6]
Ending not observed (percent of all pregnancies)	6.2	5.6	6.7	6.9
Adjusted for characteristics	[6.1]	[5.8]	[6.5]	[6.8]
Insurance-billed abortion	1.3	1.1	3.5	2.7
Adjusted for characteristics	[1.3]	[1.1]	[3.5]	[2.8]

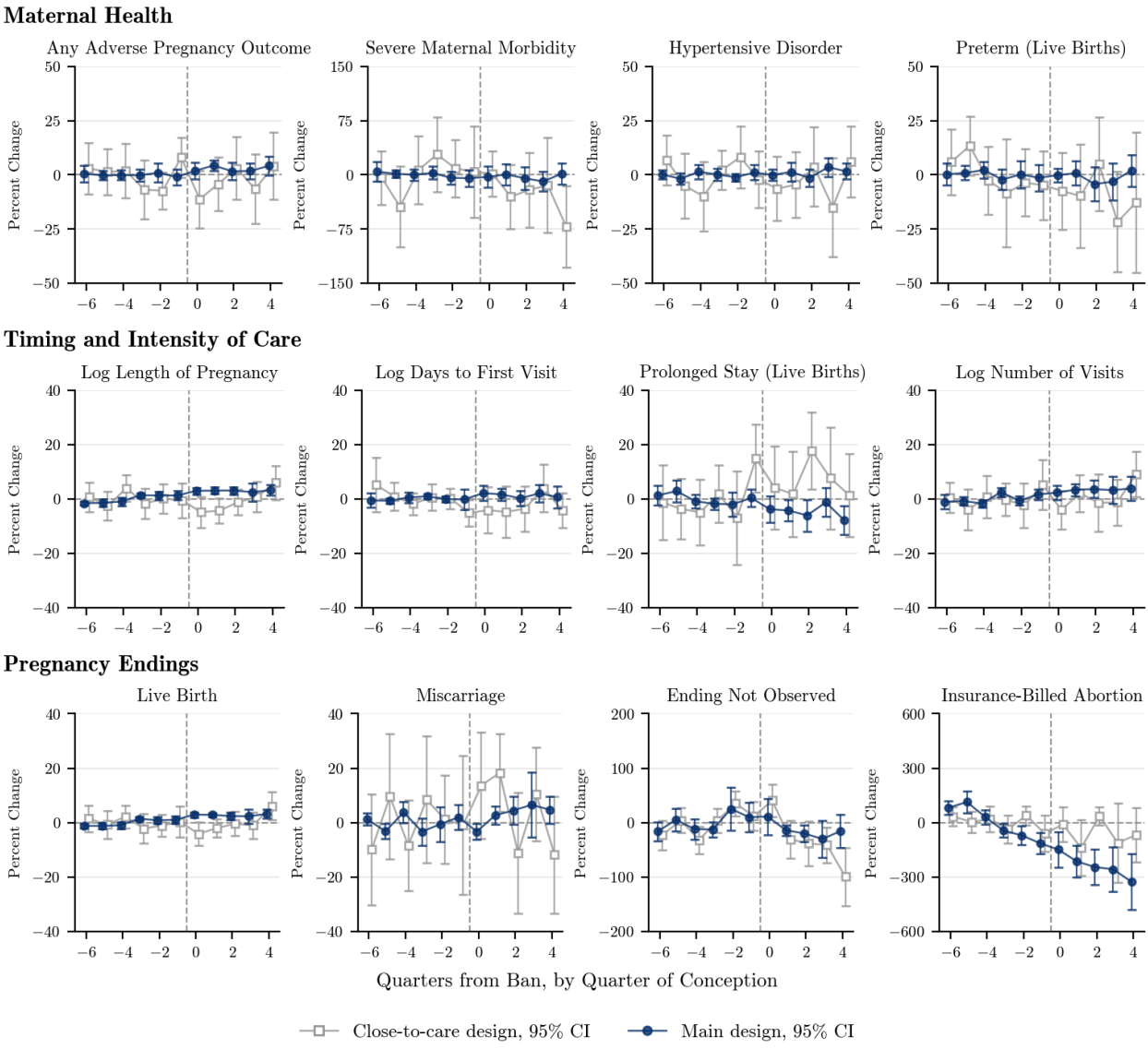
Notes: The table describes pregnancies in ban-state ZIP codes in the continuous-enrollment sample, laid out as Table 1. Beyond 50 miles are assigned ZIP codes farther than 50 miles from protected-state care (the treated group in Table A20); within 50 miles are the close-to-care ZIP codes. Before and after are relative to the ban in each pregnancy's state.

Table A20—Close-to-Care Design Estimates

Outcome	Pre-Ban Mean (1)	Controls		
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)
Maternal Health				
Any adverse pregnancy outcome	0.285	−0.009 (0.011)	−3.0	[−10.4, 4.4]
Severe maternal morbidity	0.027	−0.007 (0.004)	−26.6	[−56.9, 3.6]
Hypertensive disorder	0.210	−0.007 (0.010)	−3.2	[−12.8, 6.5]
Preterm birth (live births)	0.174	−0.016 (0.012)	−9.0	[−22.1, 4.0]
Timing and Intensity of Care				
Log length of pregnancy	5.265	−0.009 (0.015)	−0.9	[−3.8, 2.0]
Log days to first visit	3.573	−0.025 (0.035)	−2.5	[−9.3, 4.3]
Prolonged delivery stay (live births)	0.230	0.016 (0.011)	6.8	[−2.8, 16.5]
Log number of visits	2.522	0.008 (0.025)	0.8	[−4.0, 5.6]
Pregnancy Endings				
Live birth	0.785	−0.003 (0.009)	−0.4	[−2.7, 1.9]
Miscarriage	0.165	0.007 (0.008)	4.1	[−5.2, 13.3]
Ending not observed	0.062	−0.020 (0.009)	−32.4	[−59.5, −5.3]
Insurance-billed abortion	0.012	−0.007 (0.005)	−56.9	[−141.4, 27.5]

Notes: The sample is pregnancies in ban-state ZIP codes in the continuous-enrollment sample. Treated pregnancies are in assigned ZIP codes farther than 50 miles from protected-state emergency care (Section A9), and comparison pregnancies are within 50 miles. Column 1 is the treated mean in event quarters −6 to −1 in each outcome's units. Column 2 is the stacked difference-in-differences (DiD) estimate of the post-ban quarters against the average of quarters −6 to −1, comparing each cohort's treated ZIP codes only with close-to-care ZIP codes in states banned on the same date, with the Controls covariates, the weights of Wing, Freedman and Hollingsworth (2024), and standard errors (SE), clustered by assigned ZIP code, in parentheses. Column 3 is the estimate as a percent of the pre-ban mean (100 times the coefficient for outcomes in logs), and column 4 its 95 percent confidence interval (CI).

Figure A8. Event Studies, Close-to-Care and Main Designs



Notes: The figure plots stacked event-study coefficients of the close-to-care design (hollow markers), comparing ban-state ZIP codes farther than 50 miles from protected-state care with close-to-care ZIP codes in states whose ban took effect on the same date, beside the main design's coefficients from Figure 2 (filled markers), in the Controls specification and the continuous-enrollment sample. Bars show 95 percent confidence intervals from standard errors clustered by assigned ZIP code in the close-to-care design and by state in the main design. The pre-ban coefficients are normalized to average zero over quarters −6 to −1, and the dashed vertical line falls between quarters −1 and 0. The vertical axis, percent change, is each coefficient as a percent of its own design's treated ban-state pre-ban mean for shares and 100 times the coefficient for outcomes in logs, and vertical axes are drawn as in Figure 2.

Table A21—Outcome Construction

Outcome	Definition	Codes	Form
Pregnancy Endings			
Live birth	Ending type after the validated hierarchy (live birth, stillbirth, molar, ectopic, induced abortion, miscarriage), among observed endings.	CM Z37.0, Z37.2, Z37.3, Z37.5, Z37.6, O80, O82; PCS 10D00Z0–10D00Z2, 10D07Z3–10D07Z8, 10E0XZZ; CPT 59400–59622 (delivery); DRG 765–768, 774, 775, 783–788, 796–798, 805–807	Indicator
Miscarriage	Ending type after the same hierarchy, among observed endings.	CM O02.0, O02.1, O02.8, O02.9, O03; CPT 59812, 59820, 59821; PCS 10D17Z; DRG 770, 779	Indicator
Ending not observed	Two or more dates with prenatal, adverse-outcome, or Z3A codes, no gap over 60 days, and no ending code.	CM Z34, Z36, O09, Z3A; CPT 59425, 59426, 76801–76817, 59000–59015; adverse-outcome codes below	Indicator
Insurance-billed abortion	Ending type after the same hierarchy, among observed endings.	CM Z33.2, O04, O07; CPT 59840, 59841, 59850–59852, 59855–59857; PCS 10A0; HCPCS S0190, S0191, S0199, S2260, S2265–S2267	Indicator
Maternal Health (window: pregnancy start through 42 days after the ending for SMM; pregnancy otherwise)			
Any adverse pregnancy outcome	Any of infection or sepsis, preterm birth, gestational diabetes, poor fetal growth, or placental abruption. Preterm birth: live births under 259 days from start to delivery if dated by Z3A codes or embryo transfer, otherwise a preterm-delivery diagnosis.	Infection CM A40, A41, A32.7, R65.20, O85, O86, O41.12, T80.211A, T81.4XXA, T81.44; preterm dating CM Z3A, CPT 58974, 58976, 89258, diagnosis CM O60.1, O60.2; gestational diabetes CM O24.4; fetal growth restriction CM O36.5; abruption CM O45	Indicator
Severe maternal morbidity	Any of the 16 CDC diagnosis indicators on any claim; the five procedure indicators (transfusion, hysterectomy, ventilation, temporary tracheostomy, conversion of cardiac rhythm) are not used.	Leonard et al. (2020, Appendix 1)	Indicator
Hypertensive disorder	Diagnosis.	CM O10–O16	Indicator
Preterm birth	Live births only: under 259 days from start to delivery if dated by Z3A codes or embryo transfer, otherwise a preterm-delivery diagnosis.	CM Z3A; CPT 58974, 58976, 89258; CM O60.1, O60.2	Indicator
Timing and Intensity of Care (window: pregnancy)			
Log length of pregnancy	Pregnancy start to ending, among observed endings of 28 to 315 days.	Start dated from CM Z3A, embryo transfer, or screening tests	ln
Log days to first visit	Pregnancy start to the first claim of any kind, among observed endings; values under 10 days set to missing.	All claims in the window	Wins. p99; ln
Prolonged delivery stay	Live birth whose delivery admission lasts more than 2 days after a vaginal delivery or more than 4 after a cesarean, the 48 and 96 hours federal law requires private plans to cover (42 U.S.C. 300gg-25).	Delivery admission, delivery-mode codes	Indicator
Log number of visits	Days with any claim.	All claims in the window.	Wins. p99; ln

Table A21—Outcome Construction (Continued)

Outcome	Definition	Codes	Form
Controls and Subgroups (window: 365 days before the pregnancy start unless noted)			
Obstetric comorbidity score	Leonard conditions present before pregnancy, weighted by the severe maternal morbidity points of Leonard et al. (2020): pulmonary hypertension 50; chronic renal disease 38; bleeding disorder 34; cardiac disease 31; HIV 30; anemia 20; gastrointestinal disease 12; asthma 11; substance use, autoimmune disease, chronic hypertension 10; pre-existing diabetes, neuromuscular disease 9; mental health 7; thyrotoxicosis 6; BMI 40 or more 5; prior cesarean 4; bariatric surgery 0. High risk: any of these conditions.	Leonard et al. (2020) list, 18 conditions, e.g. chronic hypertension CM I10, O10, O11; pre-existing diabetes CM E08–E13, O24.0, O24.1, O24.3, O24.8, O24.9, Z79.4	Score; indicator
Under 25; Medicaid	Enrollment-file age under 25; Medicaid payer group, each in the enrollment month nearest the pregnancy start.	Enrollment file	Indicator
Minority	A Black race code or a Hispanic ethnicity code, each counted once.	Enrollment file	Indicator

Notes: CM denotes ICD-10-CM diagnosis codes and PCS ICD-10-PCS procedure codes; CPT, Current Procedural Terminology; HCPCS, Healthcare Common Procedure Coding System Level II; DRG, Medicare Severity Diagnosis Related Group. Codes match by prefix, so a three-character code covers every code that begins with it. Pregnancy and ending codes draw on the Maternal and Infant Health code lists of Centers for Medicare & Medicaid Services (2023), the mapping of Ailes et al. (2016), and the Z3A codes of American Academy of Professional Coders (2024). In the Form column, Wins. p99 marks an outcome winsorized at its 99th percentile and ln its natural log.

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